

NUcare Health System

An Information Management System Proposal

Presented to the faculty of the
Department of Computer Science
NU Bacolod Incorporated

In Partial Fulfillment
of the Requirements for the Course
Information Management (CTINFMGL)

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CHAPTER 1

INTRODUCTION

In today's organizations and institutions, Information Management is a fundamental element in organizing and effectively managing information. This undergoes several procedures such as collecting, organizing, storing, and distributing data within an organization to maintain organization and accuracy. Ensuring that the data is accessible, secure from data breaches and helps in decision making leading to success in most of the Organizations.

Many users in today's setting, especially in clinics, still rely on traditional methods of storing data such as paper forms, logbooks, and spreadsheets. Paper forms and logbooks are widely used to collect information of patients, which is then manually transferred into spreadsheets. This process results in having physical records and soft copies stored on a computer.

In collecting information through paper forms and logbooks, there are instances where some fields are left blank or are difficult for the clinic admin to read. This can result in incorrect data being encoded into spreadsheets, causing delays in processing information during retrieval and updating. These issues may also lead to inconsistencies and conflicts in data accuracy between the original forms and the encoded spreadsheet data.

The NUcare Health System addresses these challenges by allowing clinic staff to manage patient records, medicine inventories, and schedules in a secure digital database. Students can book appointments, and the clinic admin can monitor, create, and update records accurately. By digitizing data management, the system reduces errors, ensures consistency, and enhances security against data breaches

1.1 Organizational Background

This section provides an overview of the organization where the proposed system will be implemented or can be used. It describes the nature of the organization and its operations. It also explains the current workflow and the type of information managed.

The proposed system will be implemented at the **University Health Services Clinic of National University Bacolod (NU)**. The clinic provides medical and dental services such as consultations, routine checkups, first aid treatment, and preventive healthcare. These services are offered to students, teachers, and other university personnel. The clinic is managed by licensed doctors, dentists, nurses, and administrative staff who ensure proper healthcare delivery within the institution.

Currently, the clinic collects and manages records using **manual forms, logbooks, and excel spreadsheet files**. Patient information, appointment schedules, and consultation details are written manually and stored in physical folders. Some reports are recorded using Microsoft Excel for documentation and tracking purposes. These records are handled by the clinic administrator and medical staff.

The clinic records various types of information including personal details, medical and dental histories, consultation notes, laboratory results, and prescribed treatments. Manual data handling makes record retrieval slow and increases the risk of errors, data loss, and duplication. Proper data management is essential to ensure accurate patient care, efficient clinic operations, and secure handling of sensitive health information. Implementing an organized data management system will significantly improve the clinic's efficiency and service quality.

1.2 Problem Statements

Specific issues and challenges are discussed in this section. It discusses the limitations of the existing method in terms of efficiency, accuracy, and accessibility of records. The problems presented serve as the basis for developing a system that can address the organization's needs.

1.2.1 Inconsistent and Unsecured Student Health Records

The clinic experiences frequent loss and inconsistency of student health records. This problem is caused by the absence of a centralized database and the reliance on manual record-keeping methods such as logbooks and paper files. Without proper tracking and systematic storage, records are prone to misplacement, duplication, and unauthorized access. As a result, the clinic encounters delays in patient care, incomplete medical histories, and reduced reliability of health information.

1.2.2 Inefficient Appointment Booking and Scheduling Process

The clinic faces difficulties in managing appointment bookings due to the lack of an automated scheduling system. Nurses have observed that some patients do not follow the proper booking process, as appointments are handled manually and without clear monitoring. This leads to overlapping schedules, missed appointments, and confusion among clinic staff. Consequently, clinic operations become inefficient, patient waiting time increases, and staff workload becomes unbalanced.

1.2.3 Poor Monitoring of Medical Supplies and Medicine Records

The clinic has limited capability to accurately track medical supplies and medicine usage. This issue arises from manual recording practices that do not provide real-time updates or systematic inventory monitoring. As a result, discrepancies in stock levels occur, leading to either shortages or overstocking of medicines. This negatively

affects the clinic's ability to provide timely treatment and increases the risk of expired or mismanaged medical supplies

1.3 System Objectives

The section explains what the project aims to achieve. The objectives define the expected functions and improvements that the system will provide compared to the current process. Specifically, the proposed system aims to:

1.3.1 Centralize and Digitize Patient Information Management

This objective aims to centralize and digitize the management of patient information within the NUcare Health System to replace the existing manual and paper-based record-keeping process. The system is designed to collect, store, and organize patient data such as personal information, medical history, physical examination records, and clinical transactions in a secure digital database. By providing a unified platform, the system minimizes the risk of missing, unreadable, or duplicated records that commonly occur when using paper forms and logbooks. This objective directly addresses issues related to data inconsistency, delayed retrieval, and human error during manual encoding.

1.3.2 Automate Clinic Operations and Appointment Scheduling

This objective focuses on automating clinic operations such as appointment booking, medical professional scheduling, and clinical transaction recording. The system allows students to request and schedule appointments digitally, while clinic administrators and medical professionals can manage availability, bookings, and service

records through the system. Automating these processes reduces dependency on manual logs and spreadsheets, improving efficiency and reducing scheduling conflicts. This objective also ensures that appointment statuses, preferred dates, and assigned medical professionals are accurately tracked within the system. As a result, clinic operations become more organized, timely, and reliable.

1.3.3 Streamline Medical Inventory and Resource Tracking

This objective aims to streamline the monitoring and management of medical supplies, medicines, and clinic resources through a centralized inventory module. The system records item details such as category, quantity, unit, and expiration date to ensure accurate tracking of available resources. By digitizing inventory management, the clinic can reduce errors caused by manual counting and spreadsheet updates. This objective also helps prevent shortages and wastage by allowing clinic staff to monitor item usage and availability in real time. Lastly, clinic operations become more efficient and better prepared to meet patient needs.

1.3.4 Maintain Comprehensive and Integrated Patient Medical Records

This objective focuses on maintaining complete and integrated patient medical records by linking physical examinations, dental transactions, clinic visits, and certifications into a unified patient record. The system ensures that all medical interactions related to a patient are properly recorded and connected, eliminating fragmented or scattered data. This integration allows medical professionals to easily review a patient's medical history for accurate assessment and diagnosis.

1.4 Significance of the Study

The importance of the proposed system and the benefits it can provide to different stakeholders are discussed in this section. The section also highlights how the proposed system can contribute to the organization or user's productivity and overall service quality.

1.4.1 Clinic Administrator

The clinic administrator is the primary stakeholder who will benefit from the implementation of the proposed system. The system enables efficient management of patient records, appointment scheduling, and clinic transactions through a centralized digital platform. This reduces manual paperwork, minimizes errors, and improves data accessibility. As a result, administrative processes become faster, more accurate, and more organized, leading to improved overall clinic operations.

1.4.2 Medical Practitioners

Medical practitioners, including doctors and dentists, are significant beneficiaries of the proposed system. The system allows them to access patient medical and dental histories, laboratory results, and previous examinations in an organized and timely manner. This improves clinical decision-making and ensures continuity of care. Consequently, the quality and efficiency of healthcare services provided to patients are enhanced.

1.4.3 Patients

Students and teachers also benefit from the implementation of the proposed system through convenient online appointment scheduling. The system reduces waiting time and eliminates the need for manual appointment booking. This provides greater accessibility and flexibility in managing healthcare visits.

1.5 Definition of Terms

This section presents key terminologies used in the proposed system to ensure clarity and consistency throughout the study. It defines system-specific terms to better understand the technical and functional components of the proposed system.

- **Appointment Booking:** Refers to the process of scheduling a medical consultation or service through the system. In NUcare, it allows students to reserve slots with clinic staff, helping organize daily operations and prevent scheduling conflicts.
- **Medicine Inventory:** The management and tracking of medicines, including stock levels and expiration dates. In the system, it helps clinic staff monitor supplies, alerting them when items are low or about to expire to maintain proper inventory.
- **Patient Record:** A digital collection of a student's or patient's personal information, medical history, and consultations. In NUcare, it ensures accurate documentation, making it easier for staff to track treatments and appointments.
- **Prescription Management:** The creation, storage, and retrieval of prescriptions within the system. In context, it allows clinic staff to issue prescriptions accurately and securely, reducing errors from handwritten notes.
- **Patient's Reference ID:** A unique identifier assigned to each student or patient in the system. This ID links all records, appointments, and prescriptions to the correct individual, ensuring data accuracy.

- **Schedule Monitoring:** Refers to overseeing appointments, consultations, and staff availability. In NUcare, it helps admins prevent overlapping bookings and maintain an organized clinic schedule.
- **Service Request:** A submission made by a student or patient requesting medical attention or services. The system records details such as the type of service, requested schedule, and current status for proper processing.
- **System Administrator:** A user role responsible for managing the system, including user accounts, data security, and system configuration. In NUcare, the administrator ensures smooth and secure operation of the platform.
- **User Role:** Defines the permissions and access level of each system user. In context, it ensures that staff, students, and administrators can only perform actions appropriate to their role, protecting sensitive information.

CHAPTER 2

FORMS AND DATA ANALYSIS

This chapter presents the forms and data analysis requirements of the proposed system focusing on the key information needed to support the core processes and transactions. It discusses the major input forms to be used for data entry, validation, and record maintenance, as well as the specific fields required to ensure accuracy, consistency, and completeness of stored records. Additionally, this chapter explains how the collected data will be organized, processed, and analyzed to generate meaningful outputs such as summaries, reports, and monitoring information for decision-making.

2.1 Form Description

This section presents the major forms to be used in the proposed system and explains their role in supporting the system's workflow. Each form is briefly described according to its purpose and how it will be used to automate key processes and transactions. The form descriptions also serve as a basis for designing the system interface and ensuring efficient and organized data processing.

2.1.1 Physical Examination Form

The Physical Examination Form is the primary data collection tool for NU Bacolod, used during annual health check-ups and medical clearances for students and staff. Designed to consolidate vital signs, physical assessments, and medical histories into a single digital record. This form replaces the manual, paper-based filing system, which is often prone to data loss and slow retrieval. By centralizing information, it allows nurses and physicians to record, and access data simultaneously, significantly speeding up

clinical transactions. The form includes the patient's medical results, such as high blood pressure or chronic history, for immediate physician review. The data is automatically archived into the patient's records in the database. This allows it to generate a real-time health summary and update the patient's clearance status, providing an efficient conclusion to the medical transaction while ensuring long-term health monitoring.

NU BACOLOD Health Services Office

PHYSICAL EXAMINATION FORM

ID #: ✓ **Date:** ✓

Name: ✓

Course: ✓ **Year & Section:** ✓

XRAY
YES / NO

Physical Examination

Blood Pressure: Systolic ✓ Diastolic ✓ For Blood Pressure monitoring

Pulse Rate (PR): ✓ **Height (in cm):** ✓

Weight (in kg): ✓ **Abdomen:** ✓

Ears: ✓ **Eyes, Pupils:** ✓

Heart: ✓ **Lungs:** ✓

Nose: ✓ **Skin:** ✓

Thorax: ✓ **Extremities:** ✓

Deformities: ✓

Other Pertinent Findings: ✓

Chest X-Ray

Normal ☒ For Repeat Chest X-Ray ☐

Cardio Clearance

Cleared ☒ with Significant Findings ☐

Significant Findings: ✓

Remarks: ✓

Medical History

	Yes	No
Asthma	☒	☐
Diabetes Mellitus	☒	☐
Heart Ailment	☒	☐
Hypertension	☒	☐
Kidney Diseases	☒	☐
Tuberculosis	☒	☐
History of Previous Illness/Surgical Operations:	<u>✓</u>	
Allergy:	<u>✓</u>	
Smoker:	<u>✓</u>	
Gynecological/Obstetrical:	<u>✓</u>	
Alcoholic Drinker:	<u>✓</u>	

Family Medical History

	Yes	No	Remarks
Asthma	☒	☐	<u>✓</u>
Diabetes Mellitus	☒	☐	<u>✓</u>
Heart Ailment	☒	☐	<u>✓</u>
Hypertension	☒	☐	<u>✓</u>
Kidney Diseases	☒	☐	<u>✓</u>
Tuberculosis	☒	☐	<u>✓</u>
Others	☒	☐	<u>✓</u>

Remarks: ✓

Figure 1. Physical Examination Form

2.1.2 Health Status Declaration Form (Page 1)

The Health Status Declaration Form is a screening tool used to assess a patient's current health status and potential exposure to risks before they enter a facility or participate in an activity at NU Bacolod. The first page of the form consists of questions answered with "Yes" or "No" and a checklist of potential medical symptoms. Upon submission, the system evaluates responses against safety protocols to instantly identify potential health risks.

NU BACOLOD
Health Services Office

HEALTH STATUS DECLARATION

Program Course : _____ Year&Level: _____

Name : _____ Age: _____ Sex: M / F

Complete Address: _____

Contact Number : _____ Religion : _____

Name of Parent/Guardian: _____ Contact #: _____

Are you a person with disability (PWD): YES / NO
if YES, Specify Disability : _____ present a medical certificate & photocopy of the updated PWD card

Medical History: Please Check if you have had the following diseases/illnesses

☐ Bleeding Tendencies/Disorders	☐ Heart Disease	☐ Diabetes Mellitus
☐ Vertigo / Dizziness	☐ Kidney Disease	☐ Head Injury
☐ Migraine / Severe Headache	☐ Asthma	☐ Visual Defect
☐ Epilepsy / seizure / Convulsion	☐ Mumps	☐ Hearing Defect
☐ Panic / Anxiety Attacks	☐ Chickenpox	☐ Rheumatic Heart Disease
☐ Hyperacidity / GERD	☐ Hepatitis	☐ Immunocompromised
☐ Sexually Transmitted Illnesses	☐ Scoliosis	(Ex. Cancer, Lupus, HIV, etc.)
☐ Musculo - Skeletal Injury	☐ Tuberculosis	☐ Others : _____
☐ Allergies : Food / Medication	☐ Hypertension	_____
☐ Mental Health, Behavioral Problem	☐ G6PD	_____
☐ Previous Surgical Procedure	☐ Anemia	_____

Has a doctor denied you or restricted your participation in sports or any physical activity for any reason? yes / no
If yes, why? _____

Do you have an ongoing medical condition? yes / no
Specify: _____

Are you currently taking any medication? yes / no
Specify: _____

Have you passed out or been dizzy during or after exercise? yes / no

Have you experienced / felt discomfort, pain or pressure in your chest during exercise? yes / no

Figure 2: Health Status Declaration Form (Page 1)

2.1.3 Health Status Declaration Form (Page 2)

The second page of the Health Status Declaration Form outlines the legal terms and conditions, including Consent to Treatment, Travel Health Declaration, and Health Status Waivers. It requires a signature from the patient or guardian to validate the submission. By submitting the form, the patient authorizes NU Bacolod to provide medical services and certifies that all declared information is truthful. The digital process improves the workflow by instantly holding the patient accountable, noting that false declarations result in disqualification, and allowing the physician to immediately review the record. Once submitted, the system archives the consent, enabling the physician to coordinate necessary activity limitations or medical precautions without delay.

CONSENT TO TREATMENT
 I/ We hereby grant permission to the nurse, physician and dentist of the National University Health Services to render me/ my son/daughter any medical and/or dental treatment that they deem necessary as part of first aid treatment especially during but not limited to emergency cases. I understand that the NU Health Services will make all possible effort to inform me in the event of such treatment in an emergency.

 Signature over Printed of Name of Parent/Guardian/Student
 Date:

TRAVEL HEALTH DECLARATION WAIVER
 Have you ever been advised not to travel by a medical doctor due to sickness or disability? Have you ever had an untoward health related experience during a travel?" Yes/No
 If YES, why? _____

 Signature over Printed of Name of Parent/Guardian/Student
 Date:

HEALTH STATUS DECLARATION WAIVER
 I/We hereby declare and certify that all information disclosed are true and accurate to the best of my knowledge. I/We understand that I/we may be liable for disqualification, for any false declaration or concealment of information. I/We hereby authorize the National University Health Services to conduct the examinations necessary for assessment.

I/We fully understands that the Health Services Office has limited facility to accommodate and attend to cases thus, should I/ my son/daughter have pre-existing condition, I/we shall provide an updated medical clearance and coordinate with the attending physician in terms of activity limitations/ precautions.

Furthermore, I/We hereby give my consent to the National University Health Services to release the results/examinations to the authorized office if necessary.

 Signature over Printed of Name of Parent/Guardian/Student
 Date:

Figure 3: Health Status Declaration Form (Page 2)

2.1.4 Dental Record Form(Page 1)

The Dental Record Form is used by the university dentist to document a patient's oral health history, clinical findings, and treatments. The first page features a dental chart and checklist to evaluate individual teeth and overall oral status, such as gingival inflammation or calculus. This form is utilized during check-ups and consultation. The findings of the dentists are saved to the patient's database record. This immediately updates their oral health status and flags any necessary follow-up treatments or urgent dental procedures for the medical team's attention.

MNL-ADM-HSO-F-006(01)


**NATIONAL UNIVERSITY
HEALTH SERVICES OFFICE**

DENTAL RECORD

NAME: _____ RELIGION _____ NATIONALITY _____ COURSE _____
 LAST FIRST MI TEL. NO. _____ DATE OF BIRTH _____
 HOME ADDRESS _____

OPERATION _____
 CONDITION _____

OPERATION _____
 CONDITION _____

UPPER LOWER
 LINGUAL LINGUAL

OPERATION _____
 CONDITION _____

RIGHT

OPERATION _____
 CONDITION _____

DATE OF EXAMINATION																			
AGE OF LAST BIRTHDAY																			
PRESENCE OF CALCULUS	N	Y	N	Y	N	Y	N	Y	N	Y	N	Y	N	Y	N	Y	N	Y	N
INFLAMMATION OF GINGIVA	N	Y	N	Y	N	Y	N	Y	N	Y	N	Y	N	Y	N	Y	N	Y	N
PRESENCE OF PERIODONTAL POCKET	N	Y	N	Y	N	Y	N	Y	N	Y	N	Y	N	Y	N	Y	N	Y	N
PRESENCE OF DENTO-FACIAL ANOMALY	N	Y	N	Y	N	Y	N	Y	N	Y	N	Y	N	Y	N	Y	N	Y	N
	T	P	T	P	T	P	T	P	T	P	T	P	T	P	T	P	T	P	T
NUMBER OF TEETH PRESENT																			
CARIES OF TEETH																			
CARIES INDICATED FOR FILLING																			
CARIES INDICATED FOR EXTRACTION																			
ROOT FRAGMENT																			
LOST DUE TO CARIES																			
FILLED OR RESTORED																			
TOTALS																			
FLUORIDE THERAPY																			
EXAMINER																			

Page 1

Figure 4: Dental Record Form (Page 1)

The second page of the Dental Record Form is the Treatment Record, used by the dentist to log the date, diagnosis, services rendered, and tooth location. Its purpose is to maintain a history of all clinical transactions. Upon submission, the record is saved to the database of the patient's records, providing an organized audit trail and a clear conclusion to the dental transaction.

MNL-ADM-HSO-F-006(01)

TREATMENT RECORD

DATE	DIAGNOSIS	DETAILS OF SERVICE RENDERED	LOCATION OF TEETH	CHECKED BY

Page 2

Figure 5: Dental Record Form (Page 2)

2.1.6 Medical Certificate for Dress Down Form

The Medical Certificate for Dress Down Form is issued by the university physician to grant students permission to wear alternative attire when medical conditions make the standard uniform impractical. The form specifies the validity period and recommends an alternative outfit that remains within university dress code guidelines. This process provides students with professional, verifiable proof of permission from the Health Services Office.

NU BACOLOD
Health Services Office

Dress Down

MEDICAL CERTIFICATE

DATE : _____ DEPT : _____

NAME: _____

Recommendation:

Valid Until:

MEDICAL OFFICER

Figure 6: Medical Certificate for Dress Down

2.1.7 Permit to Leave Form (Page 1)

The Permit to Leave form is used to validate permission for students to leave the campus due to medical reasons. This form is issued by the clinic administration or physicians, who verify the patient's symptoms before it is sent for approval by the Discipline Officer. The form consists of two copies, one for the student and one for the Discipline Officer, and includes the specific reason for leaving, such as illness, a family emergency, or other important matters. To be valid, the form must be signed by the patient, their adviser, and the Discipline Officer. The permit only becomes official once the Discipline Officer provides the final approval.

 NU BACOLOD Discipline Office	
PERMIT TO LEAVE	
Version 2024	BCD-ACD-SDO-F-004 Page 1 of 1 - STUD'S copy
Permit No. _____ Date: _____	
Name of Student: _____ Year/Level: _____ Student ID No. _____	
Reason for Leaving: ☐ Sickness ☐ Family Emergency ☐ Important Matters – Specify: _____	
Guardian/Parent Contacted? ☐ Yes – Name of Contact: _____ ☐ No – Reason: _____	
NB: This permit is issued based on the student's request and situation. The university is not liable for any incidents after the student leaves the premises.	
_____ Student's Signature	_____ Adviser/Subject Teacher
_____ Aldrian Jay C. Dela Paz Discipline Officer	

Figure 7: Permit to Leave Form (Page 1)

2.1.8 Permit to Leave Form (Page 2)

The second page of the form features a formal correspondence from the Health Services Office, requesting authorization for the student to leave campus due to illness or other pressing matters. This section specifies the prescribed medications and home care instructions recommended by the clinical staff. Once validated with the medical officer's signature, the document serves as an official medical permit, providing the necessary clinical justification for the Discipline Officer to grant final approval for the student's departure.

ADM - HS - F - 007


**NATIONAL UNIVERSITY
HEALTH SERVICES OFFICE**

DATE: _____

To Whom It May Concern:

Please allow _____
to go home today due to _____
_____.

Prescribed Medication/s:

MEDICAL OFFICER

Figure 8: Permit to Leave Form (Page 2)

2.1.9 Medical Certificate for Absence Form

The Medical Certificate for Absence is issued by the university physician to grant students permission to be absent on the stated date of absence due to reasons. The form specifies the reason for the absence of the student. This process provides students with professional, verifiable proof of permission by signature from the Health Services Office.

NU
1900
NU BACOLOD

**NATIONAL UNIVERSITY
HEALTH SERVICES**

MEDICAL CERTIFICATE

DATE : _____ DEPT : _____

NAME: _____

REASON/S :

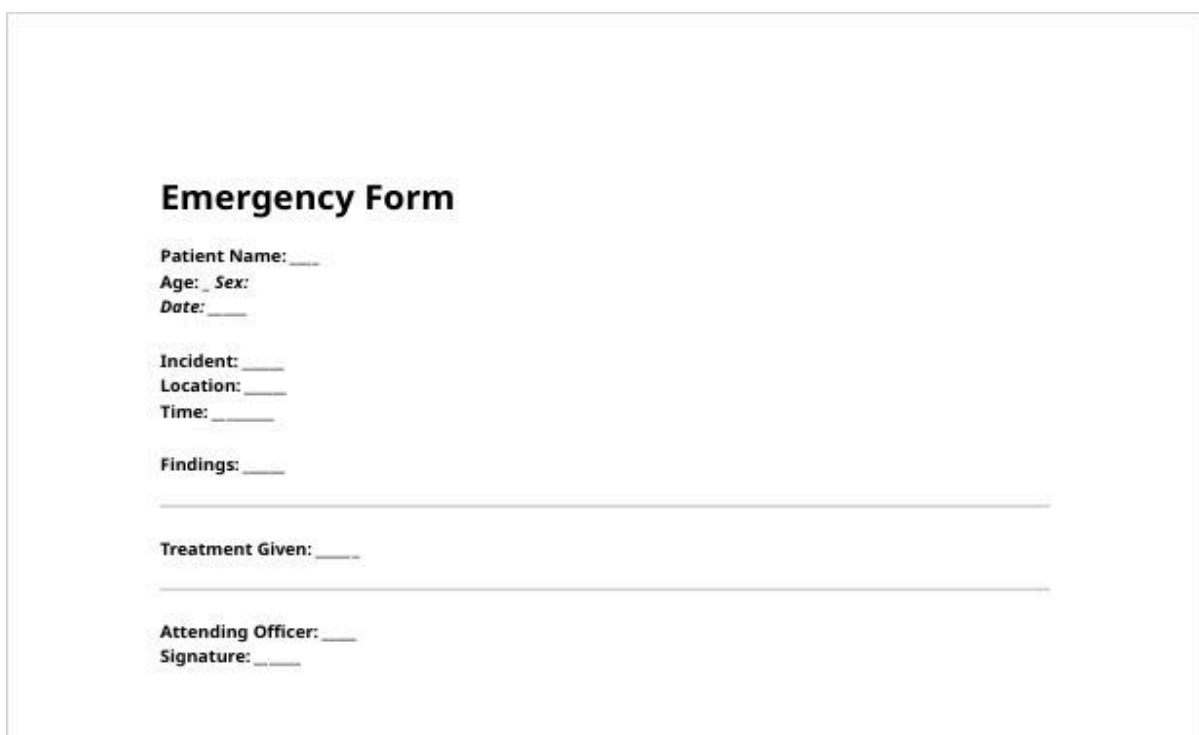
DATE OF ABSENCE:

MEDICAL OFFICER

Figure 9: Medical Certificate for Absence Form

2.1.10 Emergency Form

The Emergency Form is a critical document used to record the specific details of urgent medical incidents, including the nature of the emergency, the location, and the precise time of occurrence. It captures the physician's clinical findings and the immediate treatments administered, all of which must be validated by the attending officer's signature. This form ensures that every emergency intervention is accurately documented for the patient's permanent medical history. The system archives the record in the central database, providing an organized audit trail that is essential for long-term health monitoring and liability documentation.



The image shows a template for an Emergency Form. It is a rectangular box with a light gray border. Inside, the title "Emergency Form" is centered at the top in a bold, black font. Below the title, there are several fields for data entry, each followed by a horizontal line for the user to write. The fields are: "Patient Name:", "Age:", "Sex:", "Date:", "Incident:", "Location:", "Time:", "Findings:", "Treatment Given:", "Attending Officer:", and "Signature:". The "Findings:" and "Treatment Given:" fields have longer horizontal lines below them, indicating they are for more extensive text entry.

Emergency Form

Patient Name: _____

Age: _____ Sex: _____

Date: _____

Incident: _____

Location: _____

Time: _____

Findings: _____

Treatment Given: _____

Attending Officer: _____

Signature: _____

Figure 10: Emergency Form

2.1.11 Patient's Profile Form

The Patient's Profile Form is an essential document used to collect a patient's comprehensive personal data, including their name, date of birth, sex, nationality, religion, and contact details. It also captures critical emergency contact information to ensure rapid communication during crises. By digitizing this intake process, the clinic establishes a centralized foundation for all future medical records, replacing disorganized paper files with a secure, searchable database. This form streamlines the clinic workflow by allowing for instant data retrieval and easy updates whenever a patient visits. It enhances automation by auto-populating personal details across other medical forms, reducing repetitive data entry and minimizing human error. Once submitted, the profile is archived in the system, ensuring that every patient's health history remains monitored, organized, and readily accessible for clinical staff.

BASIC INFORMATION:
PLEASE FILL UP THE FORM BELOW

ID #: DEPARTMENT COLLEGE

FIRST NAME: SEX: NATIONALITY:

MIDDLE NAME: TEL NO. RELIGION:

LAST NAME:

BIRTHDAY:

ADDRESS:

PERSON TO BE NOTIFIED IN CASE OF EMERGENCY:

NAME:

RELATION:

ADDRESS:

TEL #:

Submit

Figure 11: Patient's Profile Form
2.1.12 Booking Form

The Booking Form is a digital appointment tool used by students, staff, and faculty to schedule medical or dental services. It requires the patient to provide brief personal details, select their desired service, and choose a preferred date and time. By digitizing this process, the university significantly reduces paper waste and provides a faster, more efficient alternative to walk-in scheduling. This form improves the clinical workflow by centralizing all requests in a single dashboard for the clinic administrator. It enhances automation by cross-referencing the requested slots with the medical staff's availability to prevent overbooking. Once the patient submits the form, the clinic administrator reviews and approves the request; the system then auto-notifies the patient of their appointment status, ensuring an organized and seamless transaction from start to finish.

PATIENT APPOINTMENT FORM

ID NUMBER:

NAME:

OCCUPATION:

EMAIL ADDRESS:

PHONE NUMBER:

SERVICE:

BOOK SCHEDULE

TIM	SUN	MO	TUE	WE	THU	FRI	SAT
8:0							
9:0							
10:							
11:0							
	LUNCH BREAK						
1:0							
2:0							
3:0							
4:0							
5:0							

SAVE

Figure 12: Booking Form

2.2 Entity-Relationship Diagram

This section presents the Entity-Relationship Diagram (ERD) of the proposed system which serves as the blueprint of how the system's data will be organized and connected. It illustrates the major entities, their attributes, and the relationships needed to support the system's transactions and record management processes. Through the ERD, the proposed database structure is clearly defined to ensure accuracy, consistency, and efficient data handling.

The purpose of the ERD, Entity-Relationship Diagram (ERD) of the **NUcare Health System** was developed to visually represent the structure of the system's database and to illustrate how data entities are related to one another. The ERD serves as a blueprint for organizing clinic data in a structured and efficient manner. It ensures that patient records, medical and dental services, scheduling, inventory, and certifications are properly stored, accessed, and maintained with minimal redundancy.

By defining entities, attributes, and relationships, the ERD supports accurate data processing, improves data integrity, and enables secure and efficient information management within the clinic environment.

The **Disease** entity stores information about medical conditions that may be diagnosed or treated within the clinic. It contains essential details such as the disease name, description, category, and relevant notes that help healthcare providers understand the nature of the illness. This entity supports accurate medical documentation and standardization of diagnoses across the system. The Disease entity

The **Patient** entity is the central component of the ERD. It stores personal and demographic information such as patient name, sex, birthdate, contact details, and address. Since most clinic activities revolve around patients, this entity is connected to several other entities in the system.

The **PhysicalExamination** entity records medical examination results, including vital signs, body system assessments, medical history, allergies, and clinical findings. This entity allows the clinic to document and track the health condition of patients during consultations.

The **Emergency** entity stores information related to emergency cases, such as incident details, vital signs, treatment given, ambulance involvement, and response times. This supports proper documentation of urgent medical situations.

The **Booking** entity manages appointment scheduling. It contains details such as service type, booking date, preferred date, and booking status. This entity ensures that patient appointments are organized and traceable.

The **Schedule** entity defines the availability of medical professionals by date and time. It supports the booking process by ensuring that appointments are made only during available time slots.

The **MedicalProfessionals** entity represents medical professionals in the clinic, including doctors and dentists. It stores their names and specializations and is linked to schedules, bookings, and certifications.

The **ClinicTransaction** entity records clinical services provided to patients, including complaints, services rendered, and items used. This entity serves as a transactional record for patient visits.

The **DentalTransaction** entity handles dental-specific records such as tooth conditions, caries indicators, fluoride therapy, diagnoses, and treatment details.

The **MedicalAndStuffs** entity stores information about medical and dental supplies, including item category, stock quantity, unit, and expiry date, supporting inventory management.

The **Program** entity represents clinic programs or departments that patients may be associated with.

The **MedicalCertificate** entity records medical certificates issued to patients, including certificate type, recommendations, and validity period.

The **PatientRecord** entity acts as a linking entity that consolidates patient information, physical examinations, and clinic transactions for organized record management.

The relationship between patients and entities such as PhysicalExamination, Emergency, Booking, ClinicTransaction, DentalTransaction, and MedicalCertificate is one-to-many, since a single patient may have multiple visits, examinations, and records over time. These relationships allow the system to maintain a complete medical history for each patient.

The **MedicalProfessionals** entity has one-to-many relationships with Schedule, Booking, and MedicalCertificate, enabling the system to manage professional availability, appointment assignments, and issued certifications.

The relationship between Booking and Schedule ensures that appointments are made based on the availability of medical professionals, preventing scheduling conflicts.

The **ClinicTransaction** entity is related to the Item entity to track supplies used during medical or dental services, supporting accurate inventory monitoring.

The **ERD** uses junction (associative) tables to resolve many-to-many (M:N) relationships. For example, a clinic transaction may involve multiple items, and a single item may be used in multiple transactions. The associative table stores the transaction ID, item ID, and quantity used. This approach allows flexible data relationships while maintaining database integrity. Similarly, junction tables support relationships such as patients enrolled in multiple programs, ensuring accurate data representation without duplication.

The **Normalization** Benefits, database design follows normalization principles to reduce redundancy and avoid data anomalies. By separating data into related entities and using foreign keys, the system ensures data consistency, improves storage efficiency, and supports reliable updates and transactions. Normalization is essential in healthcare systems where accuracy and data integrity are critical

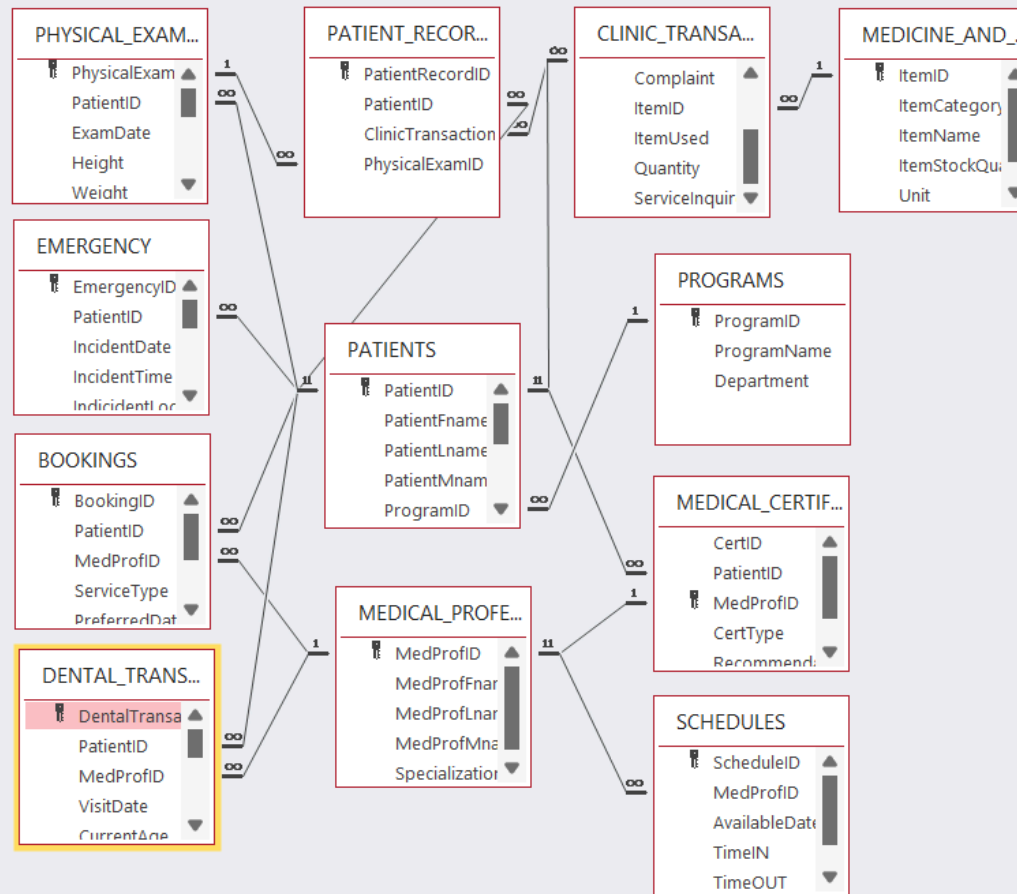
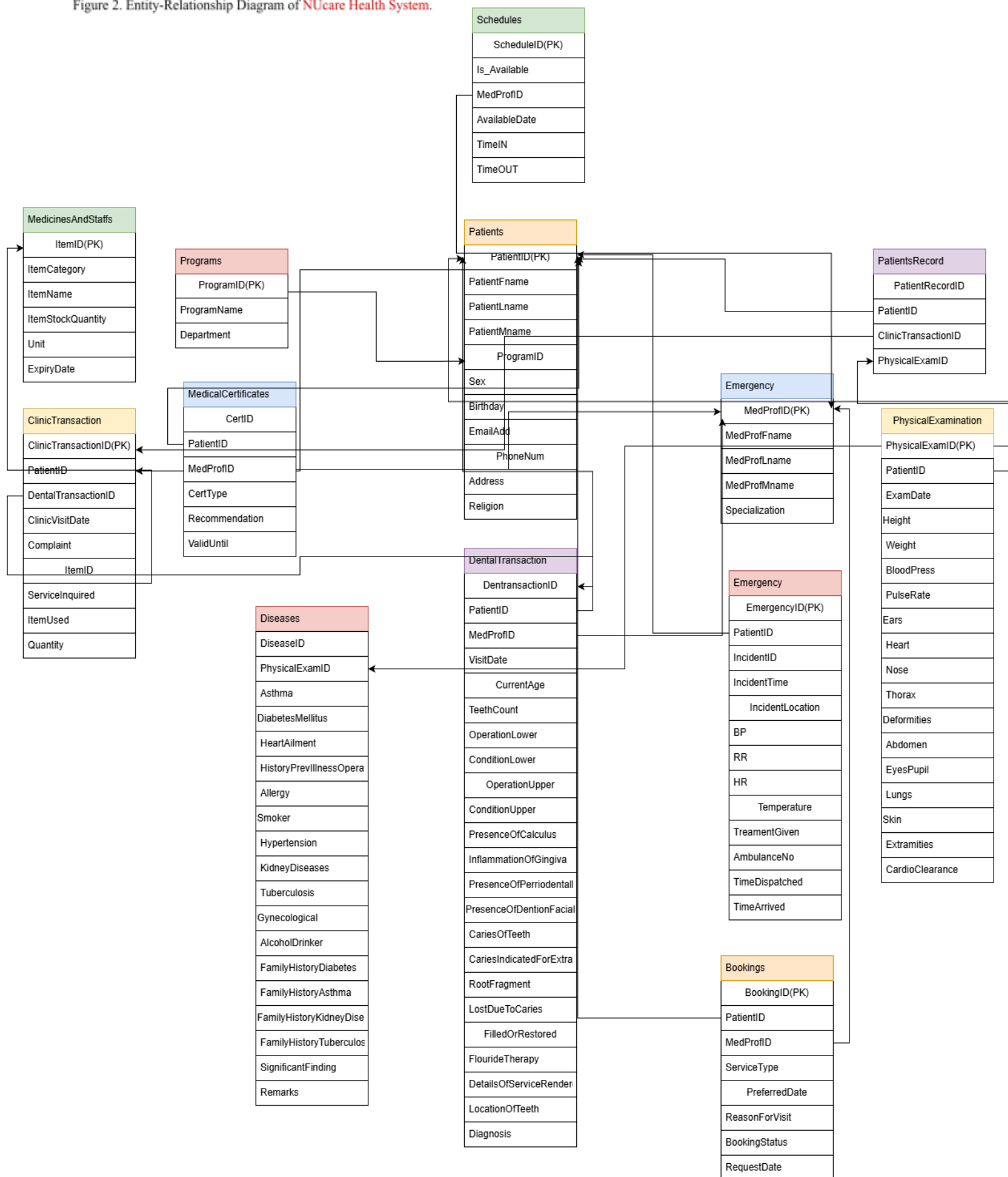


Figure 2. Entity-Relationship Diagram of **NUcare Health System**.

system, which serves as a detailed reference for all data fields to be stored and managed in the database. It defines the structure of each entity by specifying the field names, data types, formats, and descriptions to ensure consistency and accuracy in data handling. Through the data dictionary, the system's database design is clearly documented to support proper implementation, validation, and future maintenance.

The Patients entity serves as a fundamental architectural component of the clinic's database, functioning as a centralized repository for comprehensive personal and demographic data captured via the Patient's Profile Form. Managed exclusively by the Clinic Admin, this entity is strategically designed to establish a unique identity for every individual entering the system, thereby ensuring high-integrity record-keeping and facilitating highly personalized care by consolidating essential contact and background details. The operational flow revolves around two primary processes: the initial addition of new patient records and the high-frequency searching of existing profiles for clinical or administrative updates. Technically, the data structure is anchored by the PatientID, a system-generated AutoNumber primary key that guarantees uniqueness without manual input, while the ProgramID acts as a 6-character foreign key to categorize patients by their respective hospital departments. To maintain high data quality, the system enforces strict formatting rules; for instance, the PatientFname, PatientLname, and PatientMname fields are constrained to 150 characters and must be entered in Proper Case. This precision extends to demographic data like Sex (limited to binary selection), Birthday (strictly MM/DD/YYYY), and Religion, as well as communication channels including a 150-character EmailAdd, a residential Address, and a PhoneNum that must adhere to a specific 11-digit mobile string (09xxxxxxxxxx) to prevent entry errors.

Entity Name	Patients		Source Document	Patient's Profile Form
Entity Description	The Patient's Profile Form captures comprehensive personal and demographic data to establish a unique identity for every individual in the system. It ensures accurate record-keeping and facilitates personalized care by centralizing essential contact and background information.			
Process(es) Used	Add Patient record, search Patients		Triggered by	Clinic Admin
Field Name	Description	Type	Length	Format
PatientID	Unique identifier for every patient (PK)	AutoNumber	N/A	N/A
PatientFname	First name of the patient	Short Text	150	Text (Proper Case)
PatientLname	Last name of the patient	Short Text	150	Text (Proper Case)
PatientMname	Middle name of the patient	Short Text	150	Text (Proper Case)
ProgramID	Unique identifier for every program/department (FK)	Short Text	6	Text (Proper Case)
Sex	Biological sex of the patient	Short Text	150	Female Male
Birthday	Birth date of the patient	Date/Time	8	MM/DD/YYYY
EmailAdd	Personal email address of the patient	Short Text	150	name@email.com
PhoneNum	Personal phone number of the patient	Short Text	15	09xxxxxxxxx
Address	Home address of the patient	Short Text	150	Text (Proper Case)
Religion	Religion of the patient	Short Text	150	Text (Proper Case)

Table 1: Patients

The Bookings entity serves as a central operational hub for managing patient appointment lifecycles, capturing data from both the Patient's Profile and Appointment Request forms. Its primary purpose is to organize and track healthcare requests by linking them to specific patients and medical staff, supporting essential processes like record creation, status updates, and appointment monitoring. The entity's integrity is maintained by the BookingID primary key, while the PatientID and MedProfID act as foreign keys to ensure relational consistency across the database. Critical fields such as BookingStatus,

ServiceType, and PreferredDate are vital for administrative transactions, allowing the clinic to monitor workflow, organize schedules, and generate accurate output reports.

Entity Name	Bookings			Source Document	Patient's Profile Form / Appointment Request Form
Entity Description	The Bookings entity stores information related to patient appointment requests in the clinic. It records the selected service, preferred schedule, reason for visit, and current booking status. This entity enables the clinic to manage, track, and process patient appointments efficiently while linking each booking to a specific patient and medical professional.				
Process(es) Used	• Add Booking Record • Search Bookings • Update Booking Status • View Patient Appointments			Triggered by	Clinic Admin
Field Name		Description	Type	Length	Format
BookingID		Unique identifier for each booking record (Primary Key)	AutoNumber	N/A	N/A
PatientID		Unique identifier of the patient associated with the booking (Foreign Key)	Short Text	20	Text (Proper Case)
MedProfID		Unique identifier of the assigned medical professional (Foreign Key)	Short Text	20	Text (Proper Case)
ServiceType		Type of medical service requested by the patient	Short Text	100	Text (Proper Case)
PreferredDate		Date preferred by the patient for the appointment	Date/Time	N/A	MM/DD/YYYY
ReasonForVisit		Brief description of the patient's reason for consultation	Short Text	255	Text (Proper Case)
BookingStatus		Current status of the booking (e.g., Pending, Approved, Cancelled, Completed)	Short Text	50	Text (Proper Case)
RequestDate		Date when the booking request was submitted	Date/Time	N/A	MM/DD/YYYY

Table 2: Bookings

The **MedicineAndStuffs** entity, derived from Inventory Records and Medicine Stock Logs, serves as the clinic's central hub for managing medical supplies and pharmaceuticals. Managed by the Clinic Admin, it supports critical processes such as tracking stock quantities, monitoring expiration dates, and updating inventory to prevent shortages. The system uses **ItemID** as the Primary Key to ensure unique identification, while critical fields like **ItemStockQuantity** and **ExpiryDate** are essential for monitoring transactions and ensuring patient safety. By organizing data through standardized fields like **ItemCategory** and **Unit**, the entity enables the generation of accurate inventory reports and maintains consistency across the system's administrative outputs.

Entity Name	MedicineAndStuffs		Source Document	Inventory Record / Medicine Stock Log
Entity Description	The MedicineAndStuffs entity stores information about all medicines, medical supplies, and clinic-related items available in the clinic. It tracks item categories, names, stock quantities, units, and expiration dates to ensure proper inventory management, prevent shortages, and support safe patient care.			
Process(es) Used	• Add Inventory Item • Update Item Stock • Search Inventory Items • Monitor Expiry Dates		Triggered by	Clinic Admin
Field Name	Description	Type	Length	Format
ItemID	Unique identifier for each medicine or supply item (Primary Key)	AutoNumber	N/A	N/A
ItemCategory	Category of the item (e.g., Medicine, Supply, Equipment)	Short Text	100	Text (Proper Case)
ItemName	Name of the medicine or medical supply	Short Text	150	Text (Proper Case)
ItemStockQuantity	Current available quantity of the item in stock	Number	N/A	Number
Unit	Unit of measurement (e.g., pcs, bottles, boxes, tablets)	Short Text	50	Text (Proper Case)
ExpiryDate	Expiration date of the medicine or supply (if applicable)	Date/Time	N/A	MM/DD/YYYY

Table 3: MedicineAndStuffs

The MedicalProfessionals entity, sourced from the Medical Professional Information and Staff Record Forms, serves as the registry for all licensed personnel within the clinic. Its primary role is to maintain professional identities and department assignments to support patient consultations and service delivery, enabling the Clinic Admin to add, search, and assign staff to specific bookings. The entity is anchored by the MedProfID (Primary Key), an AutoNumber field that uniquely identifies each staff member, which is critical for linking professionals to the Bookings entity to monitor care assignments. Vital fields such as MedProfFname, MedProfLname, and Unit utilize a standardized Proper Case format to ensure that administrative outputs, such as staff schedules and duty logs, remain consistent and organized.

Entity Name	MedicalProfessionals		Source Document	Medical Professional Information Form / Staff Record Form
Entity Description	The MedicalProfessionals entity stores information about licensed medical personnel working in the clinic, such as doctors, nurses, and other healthcare providers. It maintains professional identity and assignment details to support patient consultations, bookings, and medical services.			
Process(es) Used	• Add Medical Professional Record • Update Medical Professional Information • Search Medical Professionals • Assign Medical Professionals to Bookings		Triggered by	Clinic Admin
Field Name	Description	Type	Length	Format
MedProfID	Unique identifier for each medical professional (Primary Key)	AutoNumber	N/A	N/A
MedProfFname	First name of the medical professional	Short Text	150	Text (Proper Case)
MedProfLname	Last name of the medical professional	Short Text	150	Text (Proper Case)
MedProfMname	Middle name of the medical professional	Short Text	150	Text (Proper Case)

Unit	Assigned department or unit of the medical professional	Short Text	100	Text (Proper Case)
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Table 4: MedicalProfessionals

The PatientRecords entity serves as a historical repository for clinical data sourced from the Patient Medical Record and Physical Examination Form. Managed by the Clinic Admin, it supports the system by tracking medical encounters and ensuring continuity of care through processes like viewing histories and linking records to transactions. The entity is anchored by the PhysicalExamID primary key, with PatientID and ClinicTransactionID acting as foreign keys to connect clinical data to specific individuals and visits. These critical fields are essential for organizing records and generating longitudinal health outputs, maintaining full consistency with the system's relational architecture.

Entity Name	PatientRecords	Source Document	Patient Medical Record / Physical Examination Form	
Entity Description	The PatientRecords entity stores clinical and examination-related information associated with a patient’s visit to the clinic. It serves as a historical record of patient consultations and physical examinations, allowing the clinic to track medical encounters and support continuity of care. Each record is linked to a specific patient and clinic transaction..			
Process(es) Used	● Add Patient Medical Record ● View Patient Medical History ● Search Patient Records ● Link Records to Clinic Transactions	Triggered by	Clinic Admin / Medical Professional	
Field Name	Description	Type	Length	Format
PhysicalExamID	Unique identifier for each physical examination record (Primary Key)	AutoNumber	N/A	N/A
PatientID	Unique identifier of the patient associated with the record (Foreign Key)	Short Text	20	Text (Proper Case)
ClinicTransactionID	Unique identifier of the related clinic visit or transaction (Foreign Key)	Short Text	20	Text (Proper Case)

Table 5: PatientRecords

The PhysicalExamination entity serves as a central clinical repository within the healthcare information system, specifically designed to capture and store the results of a patient's physical assessment. Its primary purpose is to provide a structured record of vital signs and systemic findings, which are essential for supporting diagnosis, informing treatment decisions, and maintaining a chronological history of patient health. The entity enables medical professionals to monitor health trends over time and ensures that all clinical observations are organized, searchable, and accessible for future consultations. Furthermore, this structured data entry minimizes the risk of human error by enforcing specific formats and character limits for every anatomical observation recorded. This consistency allows the system to generate standardized medical reports that can be easily interpreted by different healthcare providers across various departments. At the core of this entity's structure is the PhysicalExamID, which acts as the Primary Key to uniquely identify each specific examination instance. To maintain relational integrity within the database, the PatientID serves as a Foreign Key, linking each exam record directly to the corresponding profile in the Patient entity. This connection is vital for generating comprehensive patient reports and ensuring that medical histories remain consolidated. Beyond these identifiers, several critical fields are essential for clinical monitoring and system transactions. The ExamDate is a key field for temporal tracking, allowing the system to sort records chronologically. Quantitative fields such as Height, Weight, Blood Pressure, and Pulse Rate are fundamental for calculating health metrics and flagging potential physiological concerns. Additionally, the Cardio_Clearance field acts as a

high-priority status indicator; as a Boolean value, it provides a definitive "Yes/No" output that is often required for secondary administrative processes, such as approving a patient for sports or surgery. Together, these fields ensure that the data is not only well-organized but also actionable for generating medical clearances and clinical summaries.

Entity Name	PhysicalExamination		Source Document	Physical Examination Form
Entity Description	The PhysicalExamination entity stores the results of a patient's physical assessment conducted during a clinic visit. It records vital signs and examination findings for different body systems to support diagnosis, treatment decisions, and medical documentation. Each examination record is linked to a specific patient.			
Process(es) Used	- Record Physical Examination - Update Examination Findings - View Patient Physical Examination History		Triggered by	Clinic Admin / Medical Professional
Field Name	Description	Type	Length	Format
PhysicalExamID	Unique identifier for each physical examination record (Primary Key)	AutoNumber	N/A	N/A
PatientID	Unique identifier of the patient examined (Foreign Key)	Short Text	20	Text (Proper Case)
ExamDate	Date when the physical examination was conducted	Date/Time	N/A	MM/DD/YYYY
Height	Patient's height recorded during examination	Number	N/A	Decimal
Weight	Patient's weight recorded during examination	Number	N/A	Decimal
BloodPressure	Patient's blood pressure read	Short Text	20	Text (Proper Case)
PulseRate	Patient's pulse rate in beats per minute	Number	N/A	Number
Ears	Examination findings of the ears	Short Text	100	Text (Proper Case)
EyesPupil	Condition and response of the patient's pupils	Short Text	100	Text (Proper Case)
Heart	Heart examination findings	Short Text	100	Text (Proper Case)
Nose	Examination findings of the nose	Short Text	100	Text (Proper Case)
Thorax	Thorax examination findings	Short Text	100	Text (Proper Case)

Deformities	Noted physical deformities, if any	Short Text	100	Text (Proper Case)
Abdomen	Abdomen examination finding	Short Text	100	Text (Proper Case)
Lungs	Lung examination findings	Short Text	100	Text (Proper Case)
Skin	Skin condition findings	Short Text	100	Text (Proper Case)
Extramities	Examination findings for extremities	Short Text	100	Text (Proper Case)
Cardio_Clearance	Indicates if the patient is cleared for physical activity	Yes/No	N/A	yes no

Table 6: PhysicalExamination

The Disease entity serves as a comprehensive clinical repository within the healthcare information system, specifically designed to capture and store a patient's past and present medical conditions, lifestyle risk factors, and family medical history. Its primary purpose is to provide a structured record of health backgrounds, which is essential for supporting accurate diagnoses, risk assessments, and clinical decision-making. The entity enables medical professionals to view a patient's complete health background and ensures that significant findings are organized and accessible for ongoing care. Furthermore, this structured data entry minimizes the risk of overlooking hereditary risks or chronic conditions by utilizing a wide array of boolean flags for specific diseases. At the core of this entity's structure is the DiseaseID, which acts as the Primary Key to uniquely identify each specific illness record. To maintain relational integrity within the database, the PatientID serves as a Foreign Key, linking each medical history entry directly to the corresponding profile in the Patient entity. This connection is vital for generating comprehensive patient reports. Beyond these identifiers, several critical fields are essential for clinical monitoring and risk profiling. Fields such as Asthma, Diabetes_Mellitus, Heart_Ailment, and Hypertension are fundamental for

flagging chronic conditions that require active management. Additionally, lifestyle indicators like Smoker and Alcohol_Drinker act as high-priority risk markers; as Boolean values, they provide definitive "Yes/No" data points necessary for preventative care screening. Textual fields like Allergy and Significant_Findings allow for the documentation of specific clinical nuances that standard flags might miss. Together, these fields ensure that the data is not only well-organized but also actionable for generating risk assessments and medical clearances.

Entity Name	Disease		Source Document	Patient Medical History / Illness History Form	
Entity Description	The Disease entity stores a patient’s past and present medical conditions, lifestyle-related risk factors, family medical history, and other significant health findings. This information supports accurate diagnosis, risk assessment, and clinical decision-making. Each illness record is linked to a specific patient.				
Process(es) Used	- Record Patient Medical History - Update Illness Information - View Patient Health Background		Triggered by	Clinic Admin / Medical Professional	
Field Name		Description	Type	Length	Format
DiseaseID		Unique identifier for each illness record (Primary Key)	AutoNumber	N/A	N/A
PatientID		Unique identifier of the patient (Foreign Key)	Short Text	20	Alphanumeric
Asthma		Indicates whether the patient has asthma	AutoNumber	N/A	yes no
Diabetes_Mellitus		Indicates whether the patient has diabetes mellitus	Yes/No	N/A	yes no
Heart_Ailment		Indicates presence of heart-related conditions	Yes/No	N/A	yes no
History_Prev_Illness_Operation		History of previous illnesses or surgical operations	Short Text	255	Text (Proper Case)
Allergy		Known allergies of the patient	Short Text	255	Text (Proper Case)
Sex		Biological sex of the patient	Yes/No	N/A	yes no
Smoker		Indicates whether the patient smokes	Yes/No	N/A	yes no
Hypertension		Indicates whether the patient has hypertension	Yes/No	N/A	yes no
Kidney_Disease		Indicates presence of kidney disease	Yes/No	N/A	yes no
Tuberculosis		Indicates history or presence of tuberculosis	Yes/No	N/A	yes no
Gynecological		Indicates presence of gynecological conditions (if applicable)	Short Text	150	Text (Proper Case)

Alcohol_Drinker	Indicates whether the patient consumes alcohol	Yes/No	N/A	yes no
FamilyHistory_Diabetes	Family history of diabetes	Yes/No	N/A	yes no
FamilyHistory_Asthma	Family history of asthma	Yes/No	N/A	yes no
FamilyHistory_HeartAilment	Family history of heart disease	Yes/No	N/A	yes no
FamilyHistory_Hypertension	Family history of hypertension	Yes/No	N/A	yes no
FamilyHistory_KidneyDisease	Family history of kidney disease	Yes/No	N/A	yes no
FamilyHistory_Tuberculosis	Family history of tuberculosis	Yes/No	N/A	yes no
Other_Symptoms	Other symptoms reported by the patient	Short Text	255	Text (Proper Case)
Significant_Findings	Clinically significant findings related to illnesses	Short Text	255	Text (Proper Case)
Remarks	Additional notes or observations	Short Text	255	Text (Proper Case)

Table 7: Disease

The Programs entity serves as a fundamental organizational component of the healthcare system, designed to manage information regarding the clinic's departments and service offerings. Its primary purpose is to provide a structured reference for other entities, such as medical professionals and patient bookings, ensuring that clinical activities are properly classified for accurate institutional reporting. By standardizing these definitions, clinic administrators can efficiently add, update, or search for specific programs to maintain a clear operational hierarchy. At the core of this entity is the ProgramID, which acts as the Primary Key to uniquely identify each program or department. Critical fields such as ProgramName and Department are essential for organizing records, as they provide the descriptive labels needed for high-level system outputs and departmental filtering. These fields use a Short Text type with a 150-character limit and a Proper Case format to maintain professional consistency across all reports.

Entity Name	Programs		Source Document	Clinic Program / Department Record
Entity Description	The Programs entity stores information about the clinic's programs, services, or departments. It defines the different organizational units or service offerings within the clinic, which can be referenced by other entities such as medical professionals, bookings, or services for proper classification and reporting.			
Process(es) Used	• Add Program Record • Update Program Information • Search Programs		Triggered by	Clinic Admin
Field Name	Description	Type	Length	Format
ProgramID	Unique identifier for each program or department (Primary Key)	AutoNumber	N/A	N/A
ProgramName	Name of the clinic program or service	Short Text	150	Text (Proper Case)
Department	Department or unit associated with the program	Short Text	150	Text (Proper Case)

Table 8: Programs

The Schedules entity manages the availability of medical professionals to support appointment booking and conflict prevention. Its primary purpose is to define specific dates and time ranges when staff are available, ensuring that patient appointments are only scheduled during valid slots. This structure allows clinic administrators to efficiently manage provider shifts and maintain a clear operational timeline. The ScheduleID serves as the Primary Key, uniquely identifying each availability record, while the MedProfID acts as the Foreign Key to link schedules to specific medical professionals. Critical fields like AvailableDate, TimeIN, and TimeOUT are essential for defining shift boundaries, while the Is_Available boolean field provides real-time monitoring of slot status for booking validation. Together, these fields organize clinic workflows and ensure accurate system outputs for staff and patient scheduling.

Entity Name	Schedules	Source Document	Medical Professional Schedule / Availability Record
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Entity Description	The Schedules entity stores the availability of medical professionals in the clinic. It defines the dates and time ranges when a medical professional is available to accommodate patient appointments. This entity supports appointment booking, schedule management, and conflict prevention.			
Process(es) Used	• Add Schedule Record • Update Schedule Availability • View Medical Professional Schedules • Validate Booking Availability	Triggered by	Clinic Admin	
Field Name	Description	Type	Length	Format
ScheduleID	Unique identifier for each schedule record (Primary Key)	AutoNumber	N/A	N/A
MedProfID	Unique identifier of the medical professional (Foreign Key)	Short Text	20	Text (Proper Case)
AvailableDate	Date when the medical professional is available	Date/Time	N/A	MM/DD/YYYY
TimeIN	Start time of availability	Date/Time	N/A	hh:mm AM/PM
TimeOUT	End time of availability	Date/Time	N/A	hh:mm AM/PM
Is_Available	Indicates whether the schedule slot is available for booking	Yes/No	N/A	yes no

Table 9: Schedules

The ClinicTransaction entity is the system's operational hub, recording every patient visit and service encounter. It consolidates details like complaints, services, and supplies into a unified record to support medical history tracking and inventory management. The ClinicTransactionID is the Primary Key, with PatientID and ItemID serving as Foreign Keys to link visits to patients and supplies. Critical fields like ClinicVisitDate, Complaint, and Quantity are essential for monitoring service delivery and resource consumption, ensuring all data is organized for professional reporting.

Entity Name	ClinicTransaction	Source Document	Clinic Visit Record
Entity Description	The ClinicTransaction entity records each patient visit or service transaction in the clinic. It stores visit details such as the complaint, services inquired or rendered, and items used during the visit. This entity serves as a central reference that links patients, medical records, inventory usage, and examinations.		
Process(es) Used	• Record Clinic Visit • Search Clinic Transactions • View Patient Visit History	Triggered by	Clinic Admin

	Link Transactions to Medical Records and Inventory			
Field Name	Description	Type	Length	Format
ClinicTransactionID	Unique identifier for each clinic transaction (Primary Key)	AutoNumber	N/A	N/A
PatientID	Unique identifier of the patient involved in the transaction (Foreign Key)	Short Text	20	Text (Proper Case)
DentalTransactionID	Identifier for related dental transaction (if applicable)	Short Text	20	Text (Proper Case)
ClinicVisitDate	Date of the clinic visit	Date/Time	N/A	MM/DD/YYYY
Complaint	Primary complaint or reason for the clinic visit	Short Text	255	Text (Proper Case)
ItemID	Identifier of the medicine or supply used (Foreign Key)	Short Text	20	Text (Proper Case)
ItemUsed	Name or description of the item used during the visit	Short Text	150	Text (Proper Case)
Quantity	Quantity of the item used	Number	N/A	Number
ServiceInquired	Service requested or inquired about by the patient	Short Text	150	Text (Proper Case)

Table 10: ClinicTransactions

The DentalTransaction entity serves as the central clinical repository within the database, specifically designed to document the details of every patient encounter. Its primary purpose is to maintain a comprehensive record of a patient's oral health status, the specific procedures performed, and the professional responsible for the care at a single point in time. By centralizing this data, the entity enables the clinic to track longitudinal health trends, manage patient history, and ensure that all clinical activities are properly documented for both medical and administrative purposes. At the core of this entity's structure are the relational identifiers that integrate it into the wider system. The DentalTransactionID serves as the Primary Key, providing a unique AutoNumber reference for every visit to prevent data duplication. To link these records to the rest of the database, the PatientID and MedProfID act as Foreign Keys. The PatientID ensures that clinical findings are accurately mapped to the correct individual's history, while the

MedProfID identifies the attending dentist, which is essential for provider accountability, workload monitoring, and clinical auditing. Beyond its relational role, the entity contains several critical fields necessary for organizing records and generating meaningful outputs. Temporal and demographic data like VisitDate and CurrentAge are vital for chronological sorting and age-specific health analysis. Meanwhile, qualitative fields such as Condition_upper and Operation_lower provide the narrative detail required for dental examination forms. To facilitate rapid reporting and diagnostic monitoring, the entity utilizes "Yes/No" flags, such as PresenceOfCalculus and CariesOfTeeth, and "Number" fields like LostDueToCaries. These structured fields allow the system to automatically generate health statistics and summary reports, ensuring that the database is not just a storage tool, but an active resource for clinical decision-making.

Entity Name	DentalTransaction		Source Document	Dental Visit Record / Dental Examination Form
Entity Description	The DentalTransaction entity records dental procedures for a patient. It stores dental complaints, treatments, operations, teeth condition Each record is linked to a patient and the responsible medical professional.			
Process(es) Used	- Record Dental Visit - Update Dental Examination - View Patient Dental History		Triggered by	Clinic Admin / Dentist
Field Name	Description	Type	Length	Format
DentalTransactionID	Unique identifier for each dental transaction (Primary Key)	AutoNumber	N/A	N/A
PatientID	Unique identifier of the patient (Foreign Key)	Short Text	20	Text (Proper Case)
MedProfID	Unique identifier of the attending dentist (Foreign Key)	Short Text	20	Text (Proper Case)
VisitDate	Date of the dental visit	Date/Time	N/A	MM/DD/YYYY
Complaint	Patient's dental complaint	Short Text	255	Text (Proper Case)
CurrentAge	Patient's age at visit	Number	N/A	Number
TeethCount	Number of teeth present	Number	N/A	Number
Operation_lower	Procedures performed on lower teeth	Short Text	255	Text (Proper Case)
Condition_lower	Condition of lower teeth	Short Text	255	Text (Proper Case)
Operation_upper	Procedures performed on upper teeth	Short Text	255	Text (Proper Case)

Condition_upper	Condition of upper teeth	Short Text	255	Text (Proper Case)
PresenceOfCalculus	Indicates if calculus is present	Yes/No	N/A	yes no
InflammationOfGingiva	Indicates gum inflammation	Yes/No	N/A	yes no
PresenceOfPeriodentalPocker	indicates periodontal pockets	Yes/No	N/A	yes no
PresenceOfDentionFacialAnomaly	Indicates facial/dental anomalies	Yes/No	N/A	yes no
CariesOfTeeth	indicates teeth affected by caries	Yes/No	N/A	yes no
CariesIndicatedForExtraction	Teeth indicated for extraction due to caries	Yes/No	N/A	yes no
RootFragment	Presence of root fragments	Yes/No	N/A	yes no
LostDueToCaries	Teeth lost due to caries	Number	N/A	Number
FilledOrRestored	Teeth filled or restored	Number	N/A	Number
FlourideTherapy	Indicates fluoride therapy performed	Yes/No	N/A	yes no
DetailsOfServiceRendered	Summary of dental service rendered	Short Text	255	Text (Proper Case)
LocationOfTeeth	Teeth locations involved	Short Text	255	Text (Proper Case)
Diagnosis	Dentist's diagnosis	Short Text	255	Text (Proper Case)

Table 11: DentalTransactions

The Emergency entity documents urgent medical cases to ensure continuity of care and accurate reporting. The EmergencyID serves as the Primary Key for unique record identification, while the PatientID acts as a Foreign Key. Critical monitoring fields such as BP, RR, HR, and Temperature track physiological status. Operational fields like IncidentDate, IncidentTime, TimeDispatched, and TimeArrived are essential for organizing records and evaluating ambulance response efficiency.

Entity Name	Emergency	Source Document	Emergency Incident / Emergency Visit Form
Entity Description	The Emergency entity stores records of emergency cases handled by the clinic. It captures incident details, patient vital signs, treatment administered, and ambulance dispatch information. Each record is linked to a specific patient for continuity of care and reporting purposes.		

Process(es) Used	• Add Emergency Record • Track Ambulance Dispatch • Update Treatment Given • Search Emergency Records	Triggered by	Clinic Admin / Emergency Response Team	
Field Name	Description	Type	Length	Format
EmergencyID	Unique identifier for each emergency record (Primary Key)	AutoNum ber	N/A	N/A
PatientID	Unique identifier of the patient involved in the emergency (Foreign Key)	Short Text	20	Text (Proper Case)
IncidentDate	Date when the incident occurred	Date/Time	N/A	MM/DD/YYYY
IncidentTime	Time when the incident occurred	Date/Time	N/A	hh:mm AM/PM
IndicidentLocation	Location of the emergency incident	Short Text	255	Text (Proper Case)
BP	Blood pressure reading of the patient	Short Text	20	Text (Proper Case)
RR	Respiratory rate	Number	N/A	Number
HR	Heart rate	Number	N/A	Number
Temperature	Patient’s body temperature	Number	N/A	Decimal
TreatmentAGiven	Treatment provided during emergency	Short Text	255	Text (Proper Case)
AmbulanceNo	Ambulance vehicle number used	Short Text	20	Text (Proper Case)
TimeDispatched	Time ambulance was dispatched	Date/Time	N/A	hh:mm AM/PM
TimeArrived	Time ambulance arrived at the incident	Date/Time	N/A	hh:mm AM/PM

Table 12: Emergency

The MedicalCertificates entity records clinical certifications issued to patients, documenting the certificate type, recommendations, and validity period. Its primary purpose is to provide formal medical documentation for patient absence or work adjustments while maintaining a link to the professional who authorized the issuance. The CertID serves as the Primary Key for unique identification, while the PatientID and MedProfID act as Foreign Keys to ensure each certificate is correctly associated with a specific patient and medical professional. Critical fields such as CertType and ValidUntil are essential for organizing records and monitoring transaction outcomes, as they define

the nature of the certificate and its expiration, which is vital for generating accurate administrative outputs.

Entity Name	MedicalCertificates		Source Document	Medical Certificate Form
Entity Description	The MedicalCertificates entity stores records of medical certificates issued to patients. It records the type of certificate, the issuing medical professional, recommendations or notes, and validity period. Each certificate is linked to a specific patient and professional.			
Process(es) Used	• Add Medical Certificate • Update Certificate Details • Search Certificates		Triggered by	Clinic Admin / Medical Professional
Field Name	Description	Type	Length	Format
CertID	Unique identifier for each medical certificate (Primary Key)	AutoNumber	N/A	N/A
PatientID	Unique identifier of the patient receiving the certificate (Foreign Key)	Short Text	20	Text (Proper Case)
MedProfID	Unique identifier of the issuing medical professional (Foreign Key)	Short Text	20	Text (Proper Case)
CertType	Type of certificate issued (dressdown, absence)	Short Text	150	Text (Proper Case)
Recommendation	Notes or recommendations provided in the certificate	Short Text	255	Text (Proper Case)
ValidUntil	Expiry date of the certificate	Date/Time	N/A	MM/DD/YYYY

Table 13: MedicalCertificates

2.4 User Roles and Access Control

This section discusses the user roles and access control of the proposed system, focusing on how system access will be organized to ensure secure and efficient operations. It identifies the different types of users who will interact with the system and explains their responsibilities, permissions, and limitations based on their assigned roles.

2.4.1 User Role #1: Clinic Administrator

The Clinic Administrator is the primary user responsible for managing the overall operation of the system. This role is typically held by authorized clinic administrative personnel. Administrators are allowed to create, update, and manage patient profiles, appointment schedules, user accounts, and system-generated reports. They can also monitor the activities of other users and oversee inventory and transaction records. However, they are not permitted to modify clinical findings, medical diagnoses, or delete sensitive patient medical records without proper authorization. By controlling access and tracking all administrative actions, this role ensures accountability, proper system management, and secure auditing.

2.4.2 User Role #2: **Medical Practitioner**

The Medical Practitioner role is assigned to licensed doctors and dentists who provide healthcare services within the clinic. Users in this role are allowed to access patient medical histories, record examination results, upload laboratory findings, and prescribe treatments or medications. They can also view their consultation schedules and track patient progress over time. However, medical practitioners are restricted from deleting patient records, modifying administrative data, or granting system access to other users. Access control ensures that all medical entries are traceable to the responsible practitioner, supporting data accuracy, accountability, and proper auditing of clinical activities.

2.4.3 User Role #3: **Patients**

The Patient role is assigned to students and teachers who receive healthcare services from the clinic. Users with this role are allowed to create and manage their personal profiles, schedule appointments online, and view their consultation schedules and basic medical summaries. Patients may also access approved reports related to their health records. However, they are not authorized to access other patients' information, edit medical records, or manage administrative and clinical tasks. Role-based access control ensures that sensitive health data is protected while allowing patients to interact safely with the system, supporting secure data management and auditing.

CHAPTER 3

PROJECT TECHNICALITY

3.1 System Project Scope

This section defines the coverage of the proposed system by specifying its features, users, and included processes. It clarifies what the system will accomplish and the boundaries of the project in terms of functions, access, and implementation.

3.1.1 System Features and Modules

This section discusses the major features and modules of the proposed NUcare Health System. It explains how the system functions and how different users such as students, clinic staff, and administrators interact with the system to manage healthcare-related information efficiently. The system is designed to address issues related to manual record-keeping, data inconsistency, and delayed access to patient information.

The NUcare Health System is composed of several interconnected modules that support clinic operations and information management. Each module is designed to perform a specific function such as managing patient records, handling appointment scheduling, maintaining medicine inventories, and securing system access. Access to these modules is controlled based on user roles to ensure data security and proper system usage.

3.1.2 User Authentication and Access Control Module

The User Authentication and Access Control Module ensures that only authorized users can access the NUcare Health System. This module addresses the problem of unauthorized access and protects sensitive health information by assigning permissions based on user roles such as Clinic Administrator, Medical Staff, and Students. The Clinic

Administrator has full access to the system, Medical Staff can manage patient-related records, and Students are allowed to access limited features such as viewing personal records and booking appointments.

In this module, users are required to log in using valid credentials before accessing the system. The system verifies user identities through username and password authentication. Authorized administrators can create, update, and deactivate user accounts when necessary. The system also restricts access to specific modules depending on the assigned role. This module ensures system security, accountability, and controlled access to sensitive information.

3.1.3 Patient Records Management Module

The Patient Records Management Module is designed to digitally store and manage patient medical and dental information. This module solves the problem of misplaced paper records, unreadable handwriting, and inconsistent data caused by manual record-keeping. Clinic Administrators and Medical Staff are allowed to manage patient records, while Students can only view their own medical information.

In this module, authorized users can add new patient records using validated digital forms to ensure data completeness and accuracy. Medical staff can update patient information such as medical history, consultation notes, and treatment records. The system allows users to view and search patient records efficiently using filtering tools. Authorized users may also edit or archive records when necessary based on access permissions. This module improves data consistency and enables faster retrieval of patient information.

3.1.4 Appointment Scheduling Module

The Appointment Scheduling Module manages clinic consultation schedules in an organized and efficient manner. This module addresses scheduling conflicts, missed appointments, and manual tracking issues commonly encountered in traditional appointment systems. Students are allowed to request appointments, while Clinic Administrators and Medical Staff manage and approve schedules.

In this module, students can book appointments by selecting available dates and time slots. Clinic staff can approve, reschedule, or cancel appointments as needed. The system automatically updates appointment statuses to reflect real-time changes. Users can view scheduled appointments through the system interface. This module improves time management and ensures smooth coordination between students and clinic staff.

3.1.5 Medicine Inventory Management Module

The Medicine Inventory Management Module is responsible for monitoring and managing the clinic's medicine supplies. This module solves the problem of manual inventory tracking, stock shortages, and inaccurate medicine records. Clinic Administrators and authorized Medical Staff are allowed to access and manage this module.

In this module, users can record new medicine entries including quantity and expiration dates. Authorized staff can update stock levels after medicine distribution. The system allows users to monitor medicine availability in real time. Users can also identify low-stock or expired medicines through system notifications. This module ensures proper inventory control and supports effective healthcare service delivery.

3.1.6 Reports and Monitoring Module

The Reports and Monitoring Module provides summarized information to support clinic decision-making and documentation. This module addresses the difficulty of manually generating reports and tracking clinic activities. Only Clinic Administrators and authorized Medical Staff are allowed to generate and access reports.

In this module, users can generate reports related to patient visits, appointments, and medicine usage. The system allows users to view reports in an organized and readable format. Users can also export reports for documentation and administrative purposes. The system continuously logs activities for monitoring and auditing. This module enhances transparency and supports efficient clinic management.

3.2 Screen Interfaces

This section presents the proposed screen interfaces of NUcare Health System. It provides the key screen designed to support efficient navigation, accurate data entry, and organized management based on the system's workflow and user roles. The screen interfaces are presented to illustrate how the system will be used in actual operations and how each screen contributes to improving the overall process.

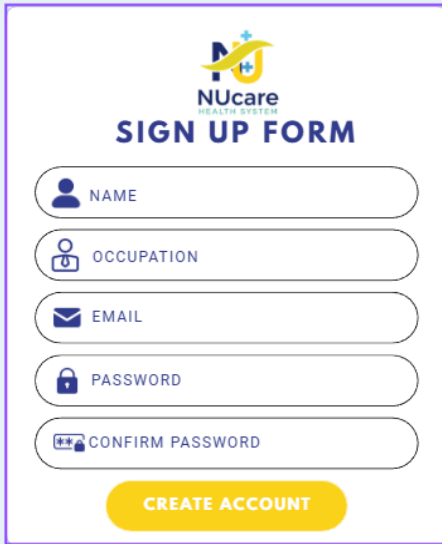
3.2.1 Login Page (Professional Side)

Figure 1 The Login Page serves as the primary secure gateway for students, faculty, staff, and medical professionals to access the system. Its purpose is to authenticate user identities, ensuring that sensitive medical records and clinic tools remain confidential and accessible only to authorized personnel. The interface consists of essential components, including Email/Username and Password input fields for credential entry, a Login button to initiate authentication, and a Sign Up button for new users. It also features a Forgot Password option for account recovery. When a user clicks the Login button, the system validates the entered credentials against the database; upon success, it redirects the user to their specific dashboard based on their role. If a new user clicks the Sign Up button, the system redirects them to the Registration Page to begin their enrollment.



3.2.2 Sign Up Page (Professional Side)

Figure 2 presents the Sign Up Page of the NUcare Health System, which serves as the registration interface for new users seeking access to the system. Its primary purpose is to gather essential personal and account information required to create a valid and secure user profile. The page contains structured input fields for Name, Occupation, Email, Password, and Confirm Password, ensuring accurate data entry and proper user identification. A Create Account button is provided to submit the registration details. Upon clicking this button, the system validates the entered information, particularly checking email uniqueness and password confirmation to ensure data consistency and security. Once validation is successful, the user's information is securely stored in the database, and the account is created. This page is designed with a clean and intuitive layout to promote ease of use while maintaining the confidentiality and integrity of user credentials.



The image shows a sign-up form for the NUcare Health System. The form is centered on a light blue background. At the top, there is a logo for NUcare Health System, which consists of a stylized 'N' and 'U' in blue and yellow, with the text 'NUcare HEALTH SYSTEM' below it. Below the logo, the title 'SIGN UP FORM' is displayed in bold, dark blue capital letters. The form contains five input fields, each with a corresponding icon on the left: a person icon for 'NAME', a person with a tag icon for 'OCCUPATION', an envelope icon for 'EMAIL', a padlock icon for 'PASSWORD', and a double padlock icon for 'CONFIRM PASSWORD'. Each input field is a rounded rectangle with a thin border. At the bottom of the form, there is a yellow button with the text 'CREATE ACCOUNT' in bold, dark blue capital letters. The entire form is enclosed in a thin purple border.

Figure 2. Sign up Page

3.2.3 Book Appointment Page (Professional Side)

Figure 3 presents the Book Appointment Page of the NUcare Health System, which functions as the primary interface for patients to schedule medical consultations within the system. Its main purpose is to guide users through a structured and secure appointment booking process by allowing them to select a clinic, choose a doctor, and specify a preferred date and time for their visit. The page displays clearly labeled sections for clinic selection, doctor selection, appointment date, appointment time, and reason for visit to ensure accurate and complete information entry. Navigation buttons such as Next and Back are provided to allow users to move sequentially through each step of the booking process, while maintaining logical flow and preventing incomplete submissions. Upon confirming the appointment, the system validates the selected schedule to ensure availability and prevent conflicts. The Book Appointment Page is designed with an organized and user-friendly layout to simplify scheduling while ensuring accuracy, reliability, and a smooth user experience.



Figure 3. Book Appointment Page(Patient Side)

3.2.4 Main Menu Page(Professional Side)

Figure 4 presents the Main Menu Page of the NUcare Health System, which serves as the central navigation hub for authorized users after successful login. Its primary purpose is to provide quick and organized access to the system's core functionalities through clearly labeled icons and buttons. The page displays menu options such as Add Patient, Visit Record, Medicine Inventory, Patient Records, Schedule Viewer, and Emergency Services, each represented by intuitive icons to enhance usability and recognition. A side navigation panel is also available, allowing users to access additional system features and return to the dashboard or log out of the system. When a user clicks any menu option, the system redirects them to the corresponding module and loads the relevant interface for that function. The Main Menu Page is designed with a clean layout and visual consistency to ensure efficient navigation, reduce user errors, and support smooth workflow within the NUcare Health System.



Figure 4. Main Menu Page (Professional Side)

3.2.5 Patient Registration Page

Figure 5 presents the Patient Registration Page of the NUCare Health System, which serves as the interface for recording new patient information into the system. Its primary purpose is to collect comprehensive personal, demographic, and contact details necessary for creating an accurate patient record. The page contains structured input fields under the Basic Information section, including ID number, department, college, first name, middle name, last name, sex, nationality, religion, telephone number, birthday, and address. An additional section titled Person to be Notified in Case of Emergency is provided to capture emergency contact details such as name, relationship, address, and contact number. A Submit button is available to finalize the registration process. Upon clicking the Submit button, the system validates all required fields to ensure completeness and correctness of the entered data. Once validation is successful, the patient information is securely stored in the database and becomes accessible to other system modules such as visit records and scheduling.

The screenshot displays the Patient Registration Page (Professional Side) with a light blue background and a dark blue sidebar on the left containing various medical icons. The main content area is divided into two sections:

BASIC INFORMATION:
PLEASE FILL UP THE FORM BELOW

Fields include:

- ID #:
- DEPARTMENT:
- COLLEGE:
- FIRST NAME:
- SEX:
- NATIONALITY:
- MIDDLE NAME:
- TEL NO.:
- RELIGION:
- LAST NAME:
- BIRTHDAY:
- ADDRESS:

PERSON TO BE NOTIFIED IN CASE OF EMERGENCY:

Fields include:

- NAME:
- RELATION:
- ADDRESS:
- TEL #:

A yellow **Submit** button is located at the bottom right of the form.

Figure 5. Patient Registration Page (Professional Side)

3.2.7 Patient Record SHS (Professional Side)

Figure 7 presents the Patient Record Page (SHS / Professional Side) of the NUcare Health System, which is designed for authorized senior high school health personnel to access and manage student patient records. Its primary purpose is to provide a centralized view of patient information relevant to SHS students, supporting accurate consultation and proper medical documentation. The page displays essential patient identification details along with recorded visit histories and medical notes specific to senior high school patients. Action buttons are provided to allow users to view detailed visit information and update medical records as needed. When a user selects a patient record or performs an action, the system retrieves the corresponding data from the database and displays it on the screen. This interface ensures efficient handling of SHS patient records while maintaining data integrity, confidentiality, and restricted access for authorized professionals.



ID NUMBER	NAME	STRAND	LEVEL & SECTION

Figure 7. Patient Record SHS Page (Professional Side)

3.2.9 Patient Record (Professional Side)

Figure 9 presents the Patient Record Page (Professional Side) of the NUcare Health System, which serves as a unified interface for authorized medical personnel to access and manage patient records across different patient categories. Its primary purpose is to provide healthcare professionals with comprehensive and organized access to patient information needed for consultation, diagnosis, and documentation. The page displays patient identification details, visit histories, and medical notes relevant to professional use. Action buttons are provided to allow users to view detailed visit records, update patient information, and add new medical entries as permitted by their access level. When a user selects a patient record or performs an action, the system retrieves the corresponding data from the database and displays it on the screen. This page ensures efficient patient record handling while maintaining data accuracy, confidentiality, and controlled access for authorized users only.

PATIENT RECORD

ID #:
 NAME:
 COURSE:
 YEAR & SECTION:

SEX:
 BIRTHDAY:
 TEL NO.

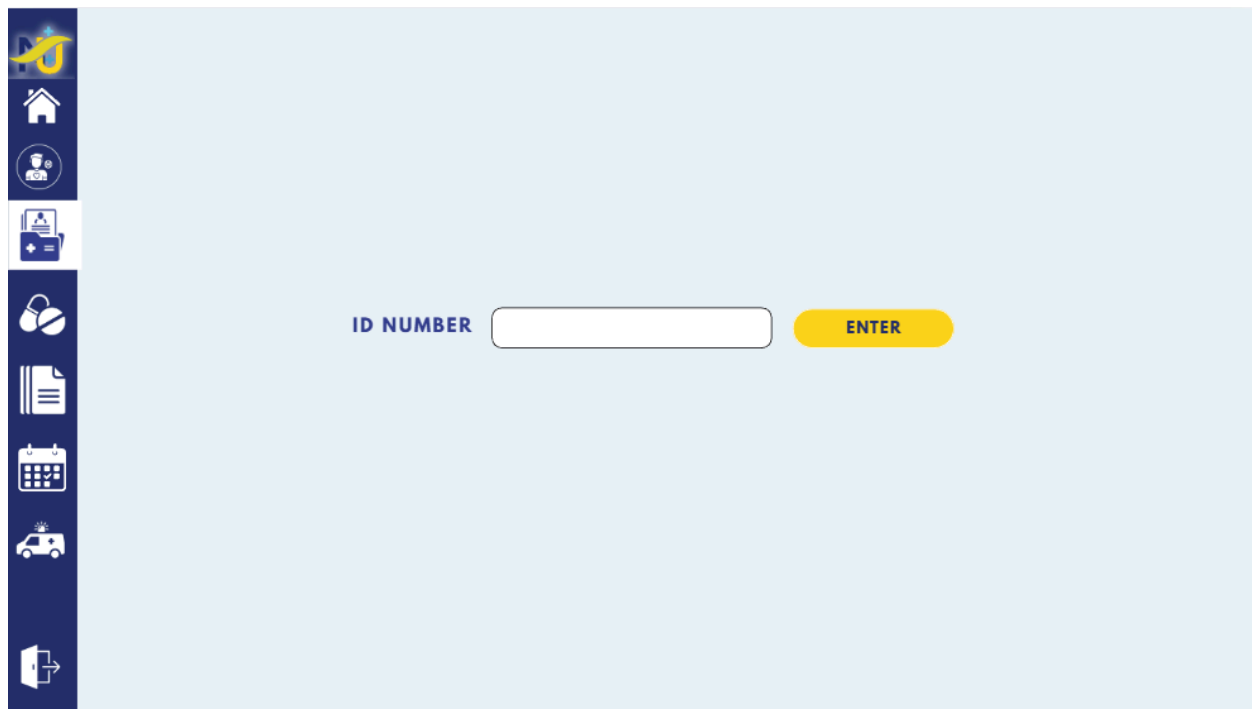
MEDICAL RECORD:

CLINIC HISTORY:

Figure 9. Patient Record Page (Professional Side)

3.2.10 Find Patient Page (Professional Page)

Figure 10 presents the Find Patient Page (Professional Side) of the NUcare Health System, designed as a focused interface for authorized healthcare personnel to locate specific patient records using unique identifiers such as patient ID. Its primary purpose is to allow professionals to quickly retrieve a patient's record by entering the patient ID into the search field. The page displays key patient information, including name, age, and basic contact details, along with action buttons to view the full profile, access visit history, or update records according to the user's access level. When a patient ID is entered, the system queries the database and immediately presents the corresponding patient record on the screen. This page ensures fast, accurate, and secure patient retrieval while maintaining data confidentiality and controlled access for professional users.



The screenshot shows a web interface for finding patient records. On the left is a vertical sidebar with a dark blue background and white icons representing different medical functions: a stethoscope, a house, a person with a heart, a clipboard with a plus sign, pills, a document, a calendar, an ambulance, and a door with an arrow. The main area has a light blue background. In the center, the text "ID NUMBER" is displayed in blue, followed by a white rectangular input field with a thin black border. To the right of the input field is a yellow button with the word "ENTER" in black capital letters.

Figure 10. Find Patient Page (Professional Side)

3.2.11 Patient Profile Page (Professional Side)

Figure 11 presents the Patient Profile Page (Professional Side) of the NUcare Health System, which provides authorized healthcare personnel with a detailed overview of a selected patient's information. This page consolidates identification details, medical history, visit records, patient concerns, prescribed medicines with their quantities, and availed healthcare services into a single, organized interface. Action buttons allow users to update patient information, add new medical entries, prescribe medicines, or record services rendered according to their access level. When a patient is selected from the Find Patient Page, the system retrieves all relevant data from the database and displays it in an easily navigable format. This page ensures accurate, secure, and efficient management of patient records while maintaining confidentiality and controlled access for professional users.

←

ID #:

NAME:

COURSE:

YEAR & SECTION:

SEX:

BIRTHDAY:

TEL NO.:

CONCERN:

MEDICINE:

QTY:

SERVICE:

SAVE

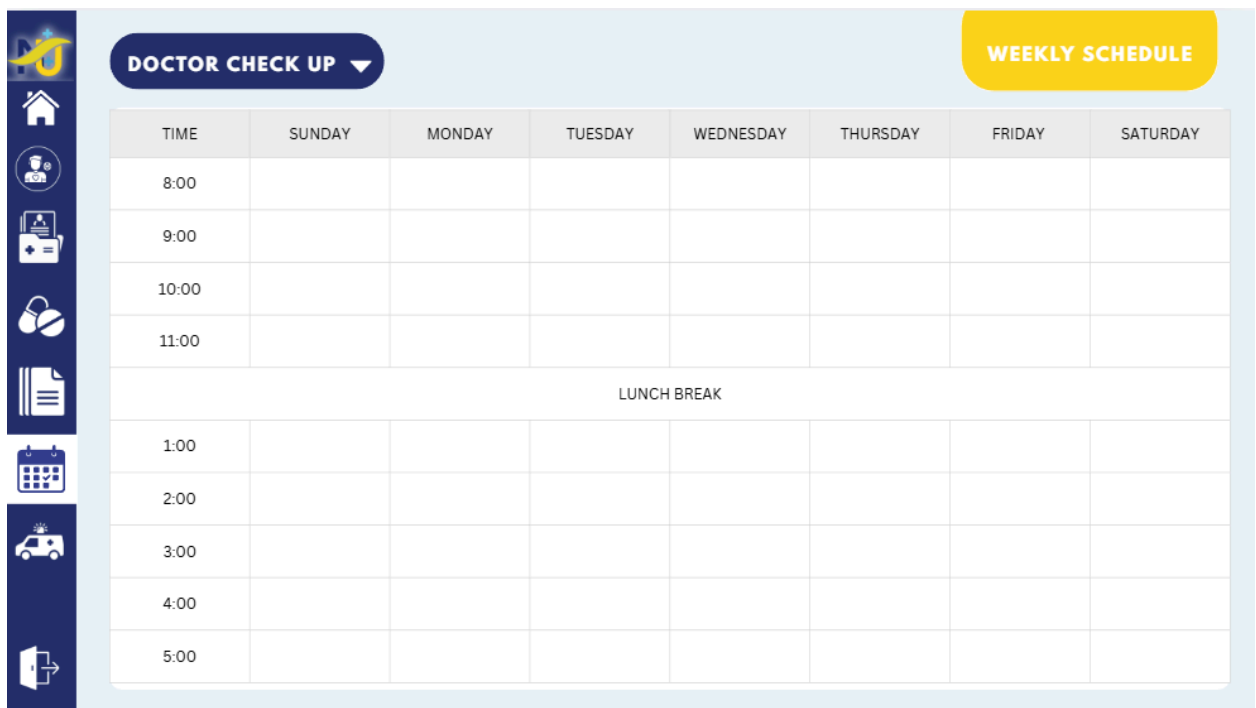
Figure 11. Patient Profile Page (Professional Side)

[illegible]

Figure 12. Summary Report Page (Professional Side)

3.2.13 Check Up Schedule (Professional Page)

Figure 13 presents the Check Up Schedule Page (Professional Side) of the NUcare Health System, designed to provide authorized healthcare personnel with a clear and organized view of upcoming patient appointments. This page displays scheduled check-ups, including patient names, appointment dates and times, assigned healthcare providers, and the purpose of each visit. Action buttons allow professionals to view detailed patient profiles, update appointment information, or reschedule as needed, depending on their access level. When a schedule is accessed, the system retrieves relevant data from the database and presents it in a chronological and easily navigable format. This page ensures efficient management of patient appointments, minimizes scheduling conflicts, and maintains accurate, secure, and controlled access for professional users.



The screenshot displays a web interface for a 'DOCTOR CHECK UP' schedule. On the left is a vertical sidebar with icons for home, user profile, appointments, prescriptions, documents, calendar, and emergency services. The main area features a header with a 'DOCTOR CHECK UP' dropdown and a 'WEEKLY SCHEDULE' button. Below is a table with columns for days of the week and rows for time slots. A 'LUNCH BREAK' row is inserted between 11:00 and 1:00.

TIME	SUNDAY	MONDAY	TUESDAY	WEDNESDAY	THURSDAY	FRIDAY	SATURDAY
8:00							
9:00							
10:00							
11:00							
LUNCH BREAK							
1:00							
2:00							
3:00							
4:00							
5:00							

Figure 13. Check Up Schedule Page (Professional Side)

3.2.14 Emergency Transport Page (Professional Page)

Figure 14 presents the Emergency Transport Page (Professional Side) of the NUCare Health System, which provides authorized healthcare personnel with a centralized interface to manage ambulance and emergency transport operations. This page displays key information such as patient details, pickup location, destination hospital, assigned ambulance, transport status, and estimated arrival times. Action buttons allow users to assign ambulances, update transport status, view patient profiles, or generate transport reports according to access permissions. When a transport request is selected, the system retrieves all relevant data from the database and presents it in real-time, ensuring timely coordination and efficient emergency response. This page ensures secure, accurate, and organized handling of patient transport while maintaining accountability and controlled access for professional users.

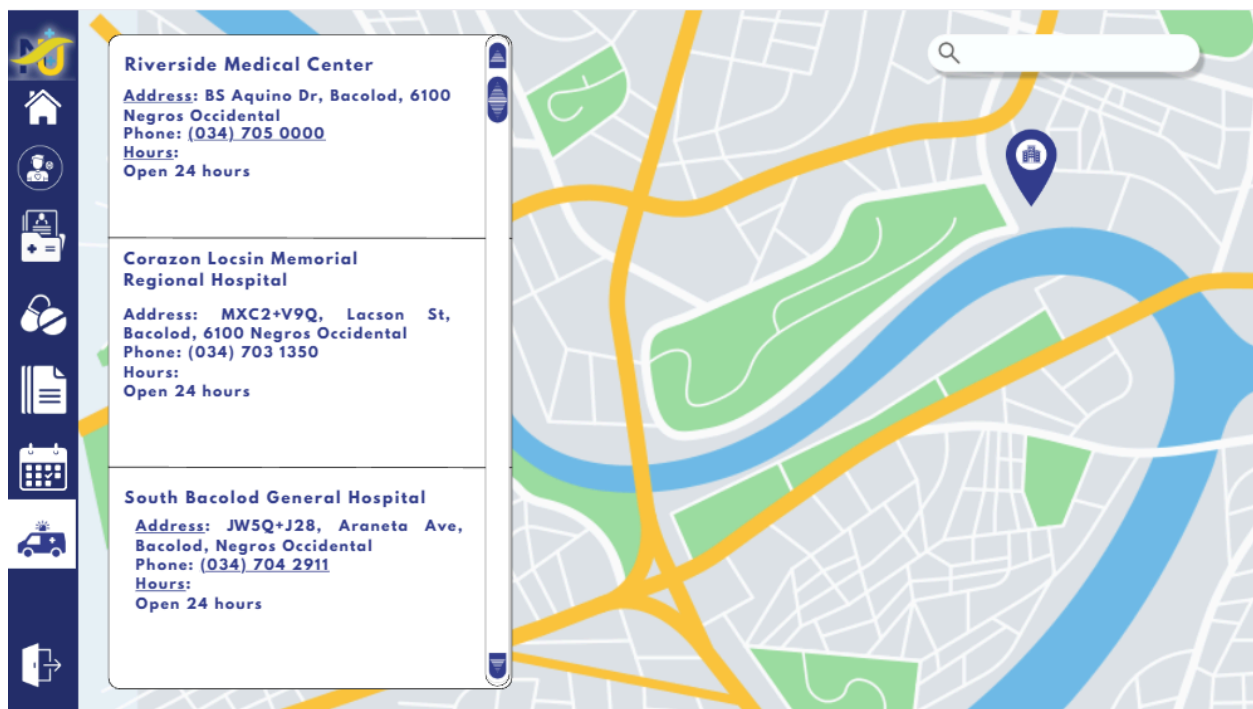
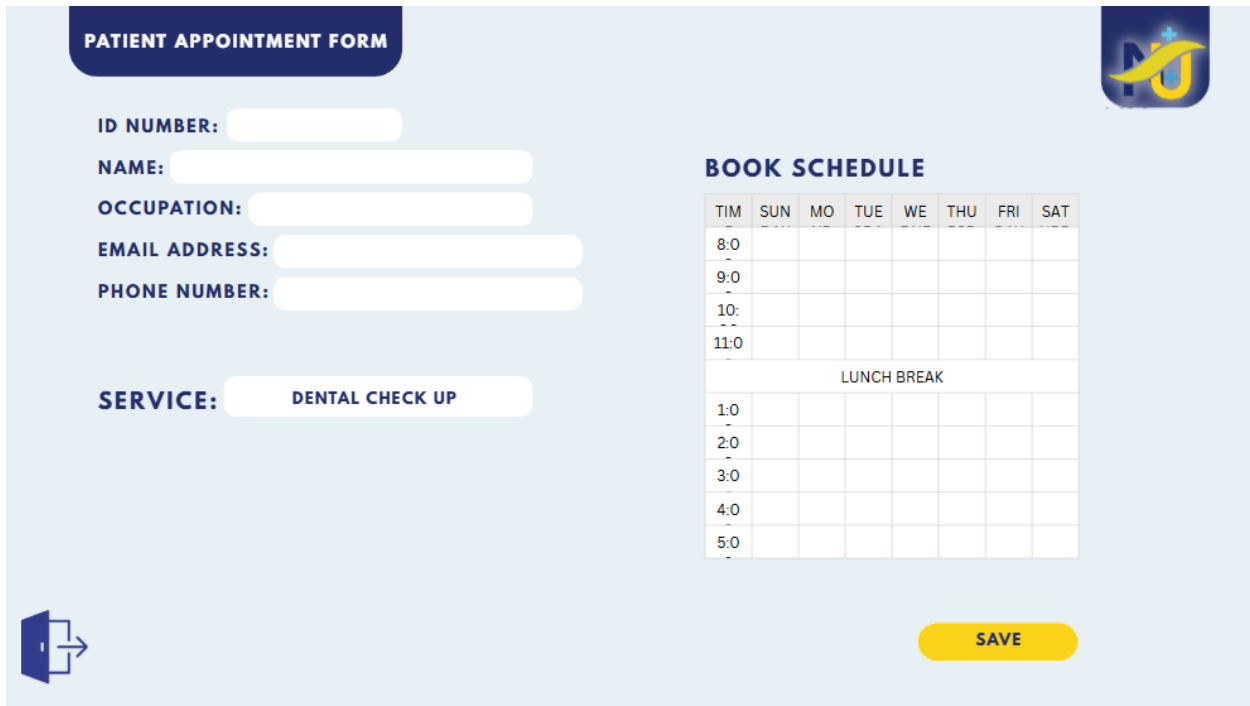


Figure 14. Emergency Transport Page (Professional Side)

3.2.15 Patient Appointment Page (Patient Side)

Figure 15 presents the Patient Appointment Page (Patient Side) of the NUCare Health System, designed to provide patients with a streamlined interface to book, view, and manage medical appointments. The page requires input of patient details such as ID, full name, occupation, email address, and phone number. Patients can also select the type of service they need and choose their preferred appointment date using an integrated booking calendar. Scheduled appointments are displayed in an organized list, showing relevant details such as service type, date, and time. Action buttons allow patients to confirm, reschedule, or cancel appointments as necessary. When a booking is submitted or updated, the system retrieves and stores the data in the database, ensuring real-time updates and accurate record-keeping. This page provides patients with convenient, secure, and organized management of their appointments while maintaining privacy and data integrity.



PATIENT APPOINTMENT FORM

ID NUMBER:

NAME:

OCCUPATION:

EMAIL ADDRESS:

PHONE NUMBER:

SERVICE: **DENTAL CHECK UP**

BOOK SCHEDULE

TIM	SUN	MO	TUE	WE	THU	FRI	SAT
8:0							
9:0							
10:							
11:0							
	LUNCH BREAK						
1:0							
2:0							
3:0							
4:0							
5:0							

SAVE

Figure 15. Patient Appointment Page (Patient Side)

3.2.16 Login Page Error Traps (Professional Side)

Figure 16 illustrates the system's handling of login errors related to user authentication. When a user enters an incorrect password, the system immediately displays an error message notifying them that the credentials provided are invalid. This prompts the user to re-enter the correct password while simultaneously preventing unauthorized access, thereby enhancing system security. Additionally, if the username or email entered does not exist in the system, an informative message alerts the user that no account is registered under those credentials and suggests creating a new account. These mechanisms not only safeguard sensitive data but also provide clear and constructive guidance to the user, reducing confusion and improving the overall login experience. By proactively managing both invalid passwords and unregistered accounts, the system ensures a secure and user-friendly authentication process while maintaining data integrity and protecting user privacy.

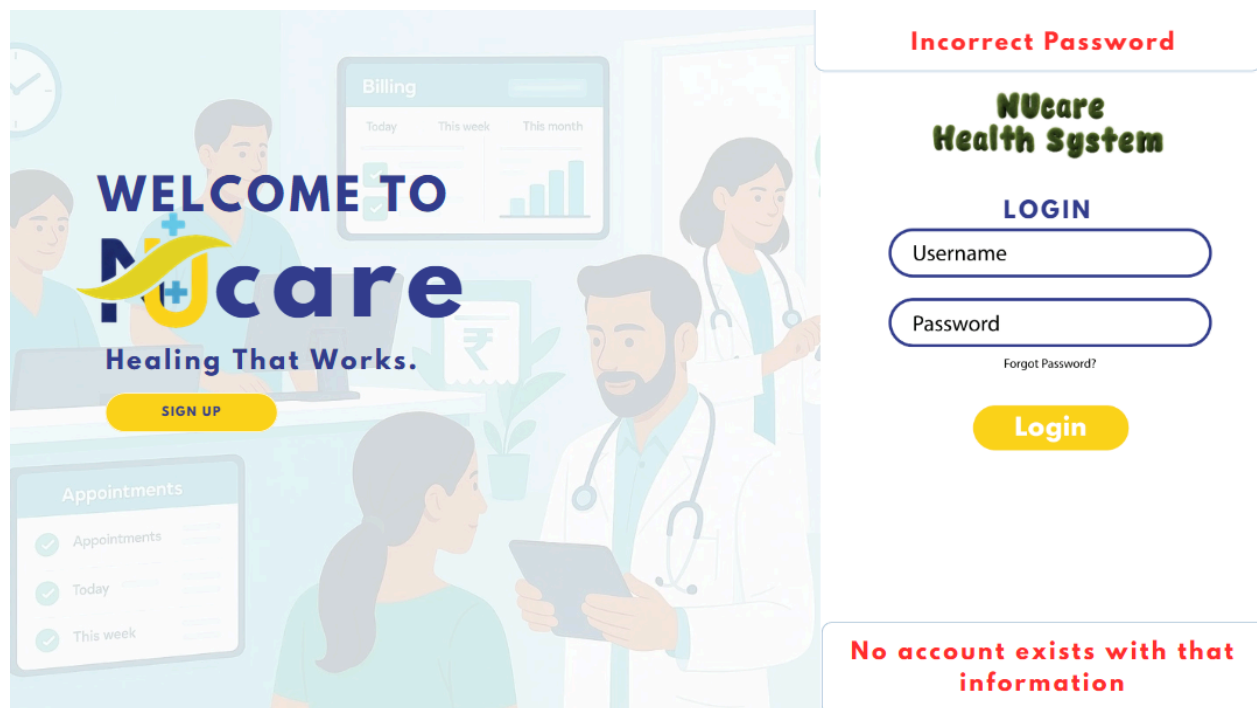
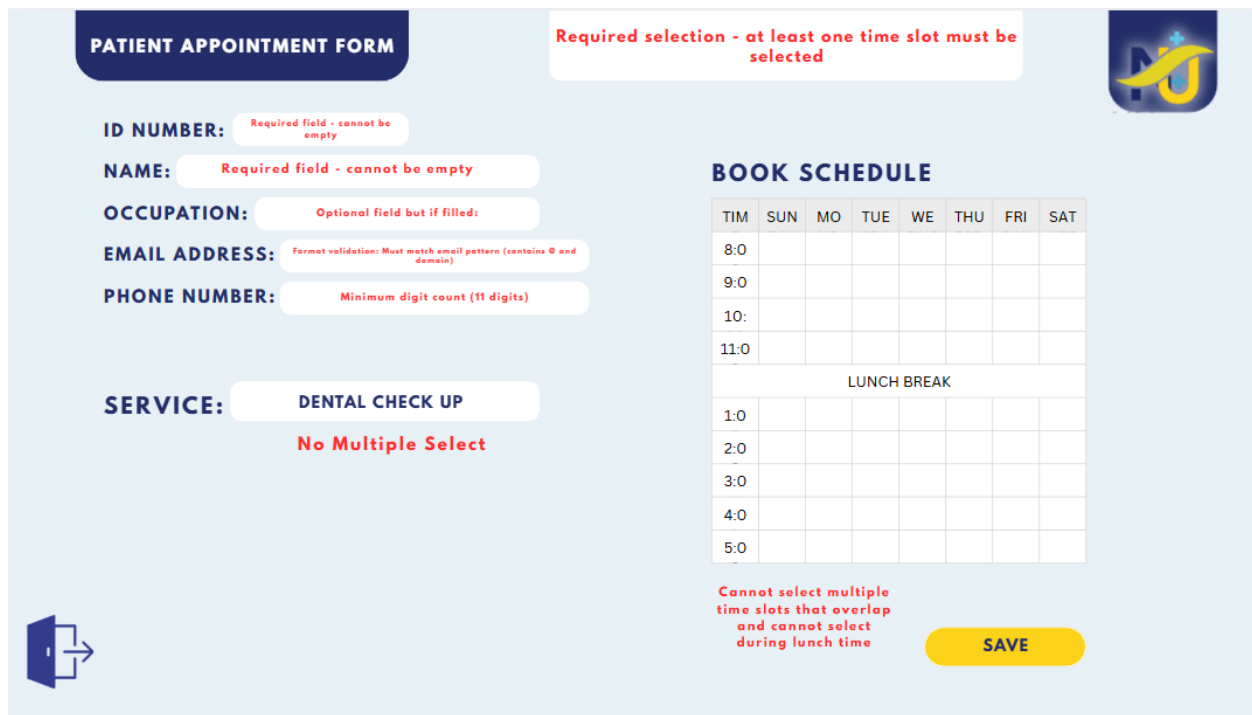


Figure 16. Login Page Error Traps (Professional Side)

3.2.17 Patient Appointment Form Error Traps (Professional Side)

Figure 17 shows the error traps incorporated in the Patient Appointment Form to ensure accurate and complete data entry. Mandatory fields such as the ID Number and Name display an error message when left empty, preventing submission until they are properly filled. Although the Occupation field is optional, any input provided is still validated to meet acceptable standards. The Email Address field enforces proper formatting by requiring a valid email pattern, while the Phone Number field restricts input to a minimum of 11 digits. The form also prevents multiple selections in the Service field, allowing only one service to be chosen at a time. In the booking schedule, users are required to select at least one time slot, and the system blocks overlapping selections as well as appointments during the designated lunch break. These validation rules act as error traps that guide users, reduce input mistakes, and ensure that only valid appointment information can be saved.



PATIENT APPOINTMENT FORM

ID NUMBER: Required field - cannot be empty

NAME: Required field - cannot be empty

OCCUPATION: Optional field but if filled:

EMAIL ADDRESS: Format validation: Must match email pattern (contains @ and domain)

PHONE NUMBER: Minimum digit count (11 digits)

SERVICE: DENTAL CHECK UP
No Multiple Select

BOOK SCHEDULE

TIM	SUN	MO	TUE	WE	THU	FRI	SAT
8:0							
9:0							
10:							
11:0							
LUNCH BREAK							
1:0							
2:0							
3:0							
4:0							
5:0							

Cannot select multiple time slots that overlap and cannot select during lunch time

SAVE

Figure 17. Patient Appointment Form Error Traps (Professional Side)

3.2.18 Emergency Transport Error Traps(Professional Side)

Figure 18 presents the error traps implemented in the hospital locator interface to ensure valid user interaction and accurate search results. The search bar includes an empty search prevention mechanism that does not allow users to perform a blank search, prompting them to enter a keyword before proceeding. The hospital list panel restricts users from selecting multiple facilities at the same time, clearly indicated by the “No Multiple Select” warning to avoid conflicting selections. These error traps help guide user behavior, prevent unnecessary system processing, and ensure that only meaningful and valid search actions and selections are performed within the interface.

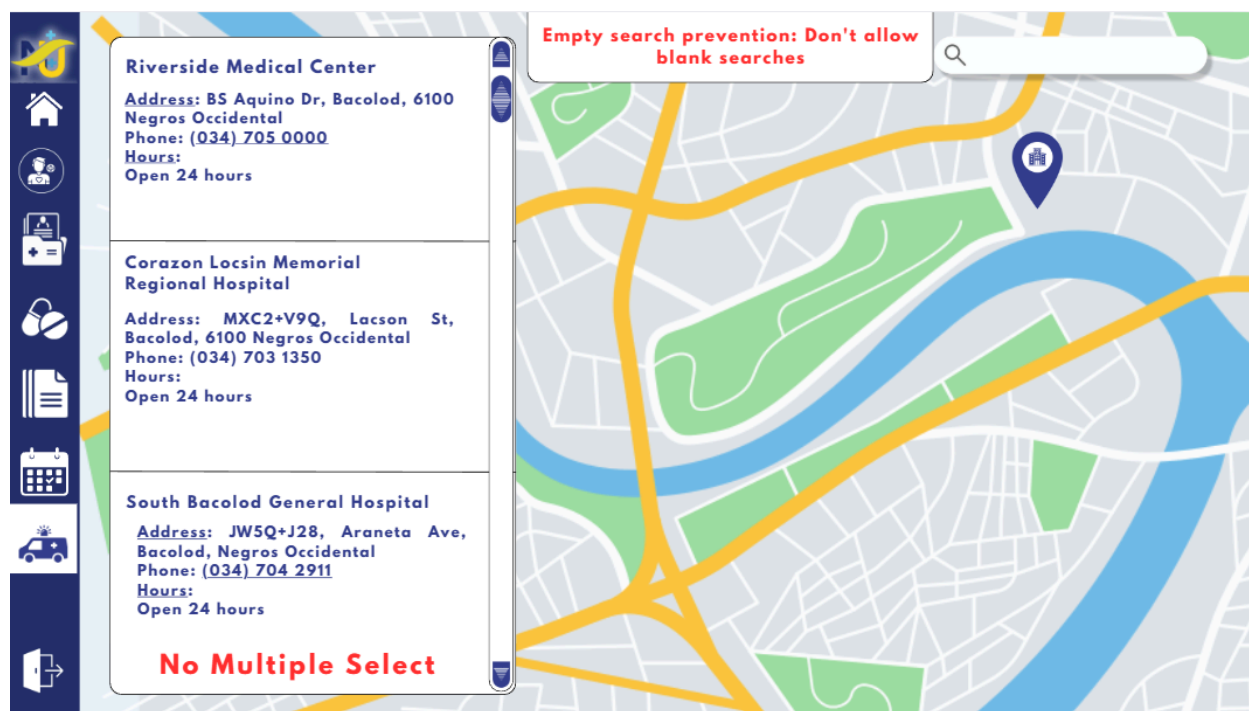


Figure 18. Emergency Transport Error Traps (Professional Side)

3.2.19 Summary Report Error Traps (Professional Side)

Figure 19 Summary Report Error Traps (Professional Side), the system incorporates robust error-handling mechanisms to ensure data integrity and prevent accidental loss of information. In this interface, users are strictly prohibited from deleting any records within the summary report, safeguarding critical data from unintended removal. Attempts to delete a record trigger an immediate error message, clearly notifying the user that such actions are not allowed. This restriction ensures that all historical and transactional data remain intact for accurate reporting, auditing, and professional reference, thereby enhancing the reliability and security of the system.



Figure 19. Summary Report Error Traps (Professional Side)

3.2.20 Patient Profile Page Error Trap (Professional Side)

As shown in Figure 20, Patient Profile Page (Professional Side), this interface allows professionals to record and manage patient information, including Course, Year & Section, ID #, Birthday, Sex, Tel No., Name, and medical details such as Concern, Medicine, Quantity (QTY), and Service. The system incorporates robust error traps to ensure accurate data entry. All required fields display the message “This field is required” if left empty, preventing incomplete submissions. The Quantity field is restricted to positive numbers only, and when a Medicine is entered, the corresponding Quantity field must also be filled to maintain consistency. These validations help safeguard patient records, reduce entry errors, and maintain the reliability and integrity of the system.

The screenshot shows the Patient Profile Page (Professional Side) with the following elements:

- Sidebar:** A vertical navigation bar on the left with icons for home, patient, and medical functions.
- Profile Section:** A header area with a back arrow, a profile icon, and a form for patient identification.

ID #:	
NAME:	SEX:
COURSE:	BIRTHDAY:
YEAR & SECTION:	TEL NO.:
- Medical Details Section:**
 - CONCERN:** A text input field with an error message: **Empty required field → "This field is required"**.
 - MEDICINE:** A text input field with an error message: **IF MEDICINE is filled, QTY must also be filled**.
 - QTY:** A text input field with an error message: **must be positive number**.
 - SERVICE:** A text input field with an error message: **Empty required field → "This field is required"**.
- SAVE Button:** A yellow button labeled **SAVE**.

Figure 20. Patient Profile Error Traps (Professional Side)

3.2.21 Find Patient Page Error Trap (Professional Side)

As shown in Figure 21, Find Patient Page (Professional Side), this interface allows professionals to search for patient records using the ID Number. To maintain accuracy and prevent invalid searches, the system incorporates error traps. If the ID Number field is left empty, an error message “ID number is required” is displayed, preventing the search from being executed. Furthermore, the system enforces a length restriction, accepting only ID numbers between 8 and 20 digits, ensuring that incomplete or incorrect entries cannot disrupt the retrieval of patient information. These validations help maintain data integrity and streamline the process of locating patient records efficiently.

The screenshot displays a web interface for finding patient records. On the left is a dark blue vertical sidebar with white icons: a magnifying glass, a house, a person, a document with a plus sign, pills, a document, a calendar, an ambulance, and a door with an arrow. The main area has a light blue background. At the top, a red-bordered box contains the text "Empty → 'ID number is required'" in red. Below this, the label "ID NUMBER" is in blue. To its right is a white input field with a red border and the text "Must be 8 - 20 digits" in red. To the right of the input field is a yellow button with the text "ENTER" in black.

Figure 21. Find Patient Page Error Traps (Professional Side)

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